

The Fermionic $U(n)^3$ Quiver: BPS Entropy, Concentration, and Chaos

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Abstract. We study a purely fermionic, $\mathcal{N} = 2$ supersymmetric gauged quantum mechanics that, as far as we can tell, has not been considered before: three bifundamental fermions A, B, C of $U(n)^3$ arranged on a triangle quiver, each in p flavours, with the cubic supercharge $Q = \sum C_{abc} \text{Tr}(A^a B^b C^c)$. It is the matrix ($D = 2$) member of the coloured models of Gurau–Witten type, so its large- n limit is planar, and it was chosen because bifundamental matter has no zero weights, which removes the mechanism that makes the BPS window of adjoint matrix models grow with the rank. In the gauge-singlet sector, exact cohomology at $(n, p) = (2, 2)$ and Lanczos diagonalisation at $(2, 3)$ show that all singlet BPS states, 90 and 1680 of them, sit at half filling, and all 90 classes at $(2, 2)$ are fortuitous. A Schur–Weyl singlet basis extends concentration to $(3, 2)$, and a semidefinite bootstrap of the singlet sector, validated against all exact data and with certificates verified in exact arithmetic, shows that the $(3, 3)$ model has no singlet BPS states in degrees $k \leq 21$, which makes all 1680 classes at $(2, 3)$ fortuitous as well; up to rank six the excluded region settles at filling fractions below about 0.27. At $p = 3$ the singlet multiplet spectrum has Gaussian-orthogonal level statistics over 10024 levels and small gaps only next to the BPS degree, while $p = 2$ turns out to be special, with a hidden $\mathcal{N} = 4$ supersymmetry on singlets and a spectrum that is not random-matrix like. We then show that the singlet index is exactly the partition function of a three-species log gas on the circle. This proves $I_0(n, 2) = (-1)^n (3n)!/(n!)^3$ and fixes the sign of the index in general, and the large- n saddle gives $\ln |I_0| = F^*(p) n^2 + O(\ln n)$ with $F^* > 0$ exactly when $p > 2$; at $p = 3$, $F^* = 1.2373$, which is 41% of the singlet log-count, and as $p \rightarrow \infty$ the BPS entropy per fermion approaches the $\mathcal{N} = 2$ SYK value $\frac{1}{2} \ln 3$. A parameter-free finite- n evaluation of the same saddle reproduces all 27 non-zero exact indices and singlet counts to within 0.6 in the logarithm. At large p and fixed n the model reduces to $\mathcal{N} = 2$ SYK with $N = 3pn^2$ fermions and coupling $J = p^2 n \langle C^2 \rangle$. We solve its Schwinger–Dyson saddle, which fixes the super-Schwarzian coefficient, $\alpha_s = 0.00844$, and predicts near-BPS thresholds $E_0(q) = 1.645 J q^2 / N$, and we test it against the exactly diagonalised $n = 1$ model: its moments approach the saddle with exact $1/p$ corrections, and every gauge-charge sector, not only the singlets, carries exactly as many BPS states as its index, distributed over the R-charge as in $\mathcal{N} = 2$ SYK. The $p \geq 3$ quiver therefore has macroscopic BPS entropy, concentration, fortuity and chaos at the sizes we can reach; concentration near half filling at $(3, 3)$ and beyond, and genuine near-BPS scaling, which the saddle shows to require $N \gtrsim 60$ fermions, are the open questions, and we describe how to attack them.

1 Introduction

The fortuity programme proposes that the BPS states responsible for the entropy of supersymmetric black holes are cohomology classes that exist at a given rank but cannot be continued to larger rank, as opposed to monotone classes that are inherited along a tower of ranks [1]. Chang, Chen, Sia and Yang argued that a theory whose BPS states are mostly fortuitous should exhibit R-charge concentration [2]: within each complex, the BPS states sit in a single degree of the R-charge. $\mathcal{N} = 2$ SYK is the paradigm [3]. There the BPS states occupy a window of width $\hat{q}$ around half filling, their number grows exponentially with the number of fermions, and the non-BPS spectrum next to them is chaotic, with the density of states of the $\mathcal{N} = 2$ super-Schwarzian [4]. Chen asked for the simplest *matrix* model with these properties, a “matrix SYK” model: purely fermionic matrix quantum mechanics with a cubic supercharge built from traces [5]. He showed that a single adjoint fermion already has fortuitous BPS states, although they spread over $N + 1$ charges, and suggested that three adjoint flavours might concentrate.

Our earlier reports [6, 7] analysed this adjoint family in detail and reached a negative conclusion. At the maximal weight the BPS window has width $W = r(w_s - 1) + 1$ for a gauge algebra of rank r , where the width $w_s \geq 2$ contributed by a single Cartan site is controlled by one bilinear form. The window therefore grows with the rank, and a concentrated large- N limit is impossible for adjoint matter at fixed degree of the supercharge. The mechanism lives on the Cartan: the adjoint representation has zero weights, and at the maximal weight the complex is a product of Cartan sites, each of which contributes its own spread in charge. This points to an obvious way out, namely to build the cubic supercharge from matter without zero weights. The simplest such matter is bifundamental, and the simplest cubic gauge invariant of bifundamentals is the trace around a triangle, $\text{Tr}(ABC)$.

This report collects what we have learned about the resulting model. We describe it in Section 2 and place it in the literature in Section 3. Section 4 presents the exact and numerical results on BPS states at small rank: singlet indices, concentration at $(n, p) = (2, 2)$ and $(2, 3)$, and a fortuity test. Section 5 turns to the non-BPS

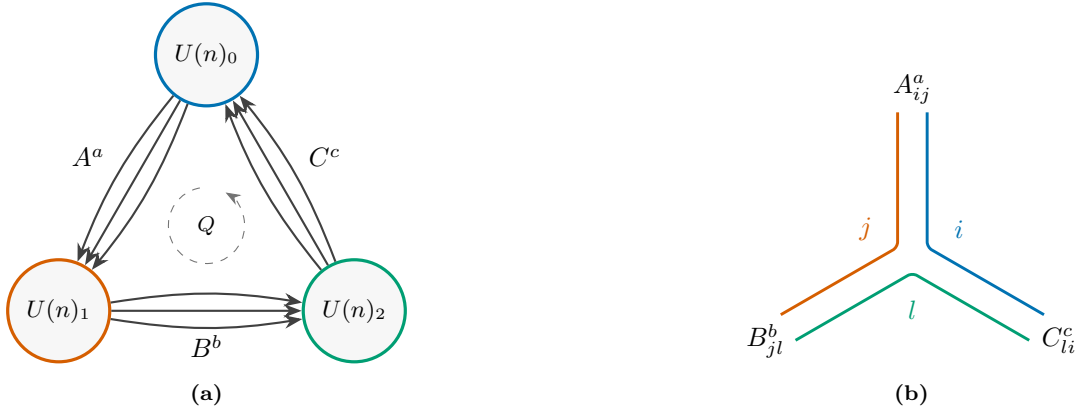


Figure 1: (a) The quiver. An arrow from node v to node $v+1 \pmod{3}$ is a bundle of p fermionic bifundamentals in $(\mathbf{n}_v, \bar{\mathbf{n}}_{v+1})$, drawn here for $p=3$. The supercharge (2) is the gauge-invariant loop around the triangle, with a general flavour tensor C_{abc} . (b) The interaction as a ribbon vertex. Each field carries the indices of the two nodes it joins, and each index line closes around the vertex. This is the $D=2$ member of the coloured models of [8]; at large n only planar ribbon graphs survive.

singlet spectrum, where $p=3$ is chaotic and $p=2$ has a hidden extended supersymmetry. Section 6 contains the main analytic result: the singlet index is the partition function of a three-species log gas, which we solve at large n . Section 7 goes beyond $n=2$, with a Schur–Weyl singlet basis at $(3,2)$ and a semidefinite bootstrap of the singlet sector at $(3,3)$, and compares fortuity in the rank and flavour directions. Section 8 solves the large- p Schwinger–Dyson saddle and tests it against the exactly diagonalised $n=1$ model. Section 9 assesses the prospects of the $p \geq 3$ model and lists the next steps.

2 The model

The gauge group is $U(n)_0 \times U(n)_1 \times U(n)_2$, with fundamental indices i, j and l for the three nodes. The matter consists of complex fermions

$$A^a_{ij} \in (\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}), \quad B^a_{jl} \in (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}), \quad C^a_{li} \in (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n}), \quad a = 1, \dots, p, \quad (1)$$

transforming as $A \rightarrow g_0 A g_1^{-1}$, $B \rightarrow g_1 B g_2^{-1}$ and $C \rightarrow g_2 C g_0^{-1}$, with canonical anticommutators such as $\{A^a_{ij}, (A^b_{i'j'})^\dagger\} = \delta^{ab} \delta_{ii'} \delta_{jj'}$. We regard A, B and C as creation operators acting on a Fock vacuum annihilated by their adjoints. The Hilbert space is the exterior algebra $\mathcal{H} = \Lambda^\bullet V$ on the $3pn^2$ modes of $V = \mathbb{C}^p \otimes [(\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}) \oplus (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n})]$, of dimension 2^{3pn^2} . The dynamics is fixed by a single supercharge,

$$Q = \sum_{a,b,c=1}^p C_{abc} \text{Tr}(A^a B^b C^c) = \sum_{a,b,c} C_{abc} A^a_{ij} B^b_{jl} C^c_{li}, \quad H = \{Q, Q^\dagger\} \geq 0. \quad (2)$$

Because Q is a product of three creation operators it acts on $\mathcal{H}$ as exterior multiplication by a fixed gauge-invariant three-form, and $Q^2 = 0$ holds identically: no condition is imposed on the couplings, which form an arbitrary tensor $C \in \mathbb{C}^p \otimes \mathbb{C}^p \otimes \mathbb{C}^p$. (The three fields are distinct, so no symmetrisation is implied, and flavour rotations $U(p)^3$ act on C .) All numerical results below use generic couplings, mostly random integers between 1 and 5 with a fixed seed, and several Gaussian real and complex sets as checks. Figure 1 shows the quiver and the interaction vertex.

The theory has a $U(1)^3$ of fermion numbers N_A, N_B, N_C , each raised by one unit by Q . Two combinations of them, $N_A - N_C$ and $N_B - N_A$, are the central charges of the gauge nodes, so gauge singlets have $N_A = N_B = N_C = m$ and total fermion number $k = N_A + N_B + N_C = 3m$. The total fermion number plays the role of the R-charge, and Q raises it by $\hat{q} = 3$. BPS states, $H\psi = 0$, are in one-to-one correspondence with Q -cohomology classes, and we write h^k for their number in degree k and $n(k)$ for the number of gauge singlets in degree k . The pairing with the top form of $\Lambda^\bullet V$ exchanges k and $3pn^2 - k$, so $n(k)$ is palindromic and $h^k = h^{3pn^2 - k}$, and half filling is $k_* = 3pn^2/2$. Non-BPS states come in Q -pairs $(\psi, Q\psi)$ in degrees k and $k+3$ with a common energy, and we label the pair by its charge relative to half filling, $q = k + \frac{3}{2} - \frac{3}{2}pn^2$, following [4]. Since $(-1)^k$ anticommutes

with Q , the singlet index

$$I_0(n, p) = \sum_m (-1)^m n(3m) = \sum_m (-1)^m h_{\text{sing}}^{3m} \quad (3)$$

is a lower bound on the number of singlet BPS states. The bound is saturated when the singlet BPS states sit in a single degree, which is what concentration means here. In $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ the BPS states fill three adjacent charges in the ratio 1:2:1 [3, 4]; gauge singlets live only in degrees $k \equiv 0 \pmod{3}$, so for even pn^2 that window contains exactly one singlet degree, half filling, and concentration of the singlet sector means that every singlet BPS state has $k = 3pn^2/2$. By Molien–Weyl, $n(t) = \sum_k n(k)t^k$ is the unitary matrix integral

$$n(t) = \int \prod_{v=0}^2 dU_v \prod_{v=0}^2 \det(1 + t U_v \otimes U_{v+1}^\dagger)^p, \quad I_0 = n(-1), \quad (4)$$

over Haar measure, with v taken mod 3. This integral is the starting point of Section 6.

Two structural remarks explain why the model was chosen and where its interesting range of parameters lies. The weights of the bifundamental modes are differences $e_i - f_j$ of weights of two different nodes and never vanish, so the zero-weight (Cartan) modes on which the window-widening mechanism of the adjoint models lives are simply absent. This removes the obstruction of [6, 7]; it does not by itself imply concentration. Second, the gauge group has dimension $3n^2$ against $3pn^2$ fermion modes, a ratio of p rather than a large number, so the singlet constraint is severe. At $p = 1$ it leaves essentially nothing: at $n = 2$ and 3 every irrep’s complex has index ± 1 , the singlet index vanishes, and concentration is vacuous. The model becomes interesting at $p \geq 2$.

3 Relation to previous work

Witten obtained the large- N physics of SYK without disorder from a coloured model of $q = D+1$ fermion fields of rank D , every pair of which shares one index [8]. For $q = 3$ and $D = 2$ the three fields are exactly bifundamentals of a three-node quiver and the vertex is $\text{Tr}(ABC)$. Witten’s melonic dominance argument, however, requires $D \geq 3$ (his eq. (3.10)); at $D = 2$ the relevant combinatorial invariant reduces to the genus of ribbon graphs and the large- n limit is planar. Our model is the complex, flavoured, $\mathcal{N} = 2$ supersymmetric version of this matrix member of the coloured family. Supersymmetric tensor versions of SYK exist with $\mathcal{N} = 1$ supersymmetry and higher-rank tensors [9]. Chang, Colin-Ellerin and Rangamani showed that $\mathcal{N} = 2$ supertensor models with dynamical bosons break supersymmetry at large N , and noted that the purely fermionic $\mathcal{N} = 2$ model with an odd-degree supercharge has no evident melonic tensor version (their eq. (6.4)) [10]. Biggs, Lin and Maldacena have since closed that gap melonically, replacing tensor contractions by $SU(2)$ $3j$ symbols [11]. The quiver is a planar counterpart: purely fermionic, cubic and $\mathcal{N} = 2$, without dynamical bosons and hence without the supersymmetry-breaking mechanism of [10].

The same three-node quiver with a closed loop and a cyclic cubic superpotential is famous in a different guise, as the quiver quantum mechanics of three-centre BPS bound states in string theory. For bosonic chiral multiplets on abelian nodes, Bena, Berkooz, de Boer, El-Showk and Van den Bleeken found exponentially many “pure-Higgs” states that are invisible on the Coulomb branch and are natural candidates for single-centre black hole microstates [12]; they also remarked that their counting formula suggests an origin in fermionic degrees of freedom. Our model is a different theory, purely fermionic and non-abelian, whose BPS states are the cohomology of a finite Fock space with no Higgs or Coulomb branch, and we claim no equivalence; a naive comparison of our $n = 1$ counts with theirs already fails.

On the fortuity side, the monotone/fortuitous dichotomy and the covering construction are due to Chang and Lin [1], and concentration as its signature in SYK to [2]. Chen constructed fortuitous states in a single adjoint matrix and used a unitary matrix integral to follow them as the rank changes [5]. Tierz computed BPS spectra of $\text{Tr}\Psi^p$ models and formulated fortuity through projections between ranks, which are cochain maps [13]; we use his definition. Choi, Choi and Kim studied fortuity in the bifundamental conifold theory and introduced “generalised monotones” dressed by baryons [14], which bears directly on our fortuity test (Section 4.3). The natural tool for large n is the quantum mechanical matrix bootstrap [15, 16, 17], which Chen already suggested for matrix SYK models; our sector-resolved version is described in [7]. A literature search in October 2026 found no previous study of the fermionic quiver itself, although absence from searches is not proof.

(n, p)	modes $3pn^2$	singlet states	$I_0(n, p)$	$\ln I_0 $ per mode	$(-1)^{pn^2/2}$
(1, 2)	6	10	-6	0.299	-
(2, 2)	24	1 274	90	0.187	+
(3, 2)	54	2 115 632	-1 680	0.138	-
(4, 2)	96	5.59×10^{10}	34 650	0.109	+
(1, 3)	9	56	0	-	$(pn^2 \text{ odd})$
(2, 3)	36	889 040	1 680	0.206	+
(3, 3)	81	4.90×10^{12}	0	-	$(pn^2 \text{ odd})$
(4, 3)	144	1.08×10^{22}	9 434 197 872	0.159	+
(1, 4)	12	346	90	0.375	+
(2, 4)	48	9.70×10^8	519 750	0.274	+
(3, 4)	108	2.99×10^{19}	199 090 719 360	0.241	+
(1, 5)	15	2 252	0	-	$(pn^2 \text{ odd})$
(2, 5)	60	1.37×10^{12}	95 866 056	0.306	+
(1, 6)	18	15 184	-1 680	0.413	-
(2, 6)	72	2.29×10^{15}	28 686 212 100	0.334	+

Table 1: Exact gauge-singlet indices and singlet counts. The $n = 1$ rows are Dixon’s sum $\sum_m (-1)^m \binom{p}{m}^3$ and the Franel numbers $\sum_m \binom{p}{m}^3$. At $p = 1$ the index vanishes for $n = 2, 3, 4$. The last column is the sign that a singlet BPS sector concentrated at half filling would give; it agrees with every non-zero entry.

4 BPS states at small rank

4.1 Singlet indices

The singlet multiplicities $n(k)$ follow from the dual Cauchy identity, $\Lambda(\mathbb{C}^p \otimes V_v \otimes \bar{V}_{v+1}) = \bigotimes_{f=1}^p \bigoplus_{\lambda} S_{\lambda} V_v \otimes S_{\lambda'} \bar{V}_{v+1}$, which reduces the singlet count to products of Littlewood–Richardson coefficients. We compute these by exact quadrature on the torus and assemble the index with a transfer matrix over $U(n)$ irreps that adds one flavour at a time,

$$I_0 = \text{Tr}[(R^{(p)})^3], \quad R^{(q)} = \sum_{\lambda} (-1)^{|\lambda|} L_{\lambda}^T R^{(q-1)} L_{\lambda}, \quad (L_{\lambda})_{\mu\nu} = c_{\mu\lambda}^{\nu}, \quad (5)$$

in exact integer arithmetic. This reaches $(n, p) = (4, 3)$ and $(3, 4)$ and reproduces every value obtained earlier by direct weight counting. Table 1 lists the results together with the total singlet counts.

Three features of the table are worth stating at once. First, the $p = 2$ entries are $(-1)^n (3n)!/(n!)^3$, a formula that we found numerically and prove in Section 6. The first few indices at $p = 3$ coincided with this pattern, $|I_0(2, 3)| = |I_0(3, 2)| = 1680$, which briefly suggested that the index depends only on np and grows only exponentially in n ; the value at $(4, 3)$, which is about 550 times larger than the $p = 2$ -type value, shows that the coincidence was accidental. Second, whenever pn^2 is even the sign of I_0 is $(-1)^{pn^2/2}$, which is what a singlet BPS sector concentrated at half filling would produce, since half filling has $m = pn^2/2$; for odd pn^2 the index vanishes identically, because the particle–hole map $m \rightarrow pn^2 - m$ pairs degrees of opposite sign. Third, at fixed n the index per mode rises with p towards the singlet entropy per mode, which already suggests that large p is the SYK-like direction.

4.2 Concentration at $(2, 2)$ and $(2, 3)$

At $(n, p) = (2, 2)$ the singlet cohomology can be computed with exact (modular) ranks. There are exactly 90 singlet BPS states, all at $k = 12$, which is half filling, so the index is saturated. At $(2, 3)$ the Fock space has 2^{36} states and the singlet sector at half filling has 300 584, which calls for a different method. We restrict to the zero-weight subspace of each degree, which contains every irrep, and to the vectors invariant under the Weyl group $(S_2)^3$, which commutes with Q and fixes singlets; this reduces the dimension by about a factor of eight. On these vectors the operator

$$P = \sum_{v=0}^2 \|E_{12}^{(v)} \psi\|^2 = \sum_v \langle \psi | J_v^2 | \psi \rangle \quad (6)$$

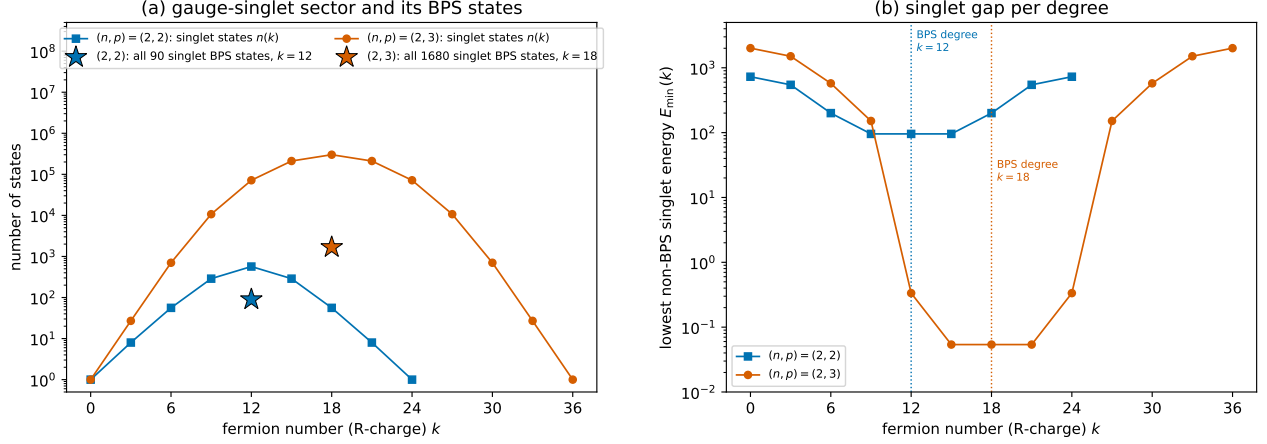


Figure 2: Concentration of the singlet BPS states at $n = 2$. (a) The number of gauge singlets in each degree k (lines) and the singlet BPS states (stars): 90 at $k = 12$ for $p = 2$ (exact ranks) and 1680 at $k = 18$ for $p = 3$ (Lanczos); there are none in any other degree. (b) The lowest non-BPS singlet energy in each degree. At $p = 3$ the gap collapses by four orders of magnitude next to the BPS degree, to 0.054 at $k = 15, 18, 21$ and 0.335 at $k = 12, 24$, while at $p = 2$ it never falls below 95.6.

is the sum of the $SU(2)$ Casimirs of the three nodes, because $J_- J_+ = J^2$ at $J_z = 0$. It commutes with H , vanishes exactly on singlets and is at least 2 on every non-singlet. The kernel of $K = H + \mu P$ is therefore the singlet BPS space, and every eigenvalue of K below 2μ is an exact singlet energy. Lanczos diagonalisation of K in the degrees $k = 9, 12$ and 15 (Weyl-symmetric dimensions 5.6×10^4 , 5.1×10^5 and 1.8×10^6), and dense diagonalisation for $k \leq 6$, finds no zero modes, with lowest eigenvalues 96.5, 0.3348 and 0.05386 at $k = 9, 12$ and 15 ; reruns at $k = 12$ and 15 with different starting vectors reproduce them to nine digits. Particle-hole symmetry excludes BPS states at $k \geq 21$ as well, so all singlet BPS states sit at $k = 18$, and by the index there are exactly $h_{\text{sing}}^{18} = |I_0| = 1680$ of them. Figure 2 summarises both cases: the singlet sector spreads over 13 degrees at $(2, 3)$, but its BPS states occupy one.

Zero-weight methods stop at $n = 2$. At $(3, 2)$ the zero-weight sectors near half filling hold 5×10^{10} states, although the singlet subspaces are only about 5×10^5 , so what is needed is an explicit basis of singlets. Section 7 describes one, and the bootstrap that takes over where even the singlet spaces are too large.

4.3 Fortuity at $(2, 2)$

Following Tierz [13], we call a rank-2 class monotone only if it lies in the image of the projection $\pi_{3 \rightarrow 2}$ that deletes every monomial involving a third index on any node. Because Q contains only creation operators, $\pi Q^{(3)} = Q^{(2)}\pi$, so the projection is a cochain map. Writing $Q^{(3)} = Q^{(2)} + Q_{\text{new}}$, a closed rank-2 state z_0 is the projection of a closed rank-3 state if and only if $Q_{\text{new}} z_0$ is exact in the subspace of states that carry index-3 modes. Grading that subspace by the number of index-3 modes, the first condition arises with two such modes, and for couplings of full rank in the summed flavour index it reduces to the vanishing of the BPS three-point function $\langle \beta | X | z_0 \rangle$ for every single letter X and every BPS state β in degree $k + 1$. A non-vanishing value obstructs the lift. By gauge covariance it suffices to test one component of each letter. At $(2, 2)$, each of the six letters obstructs a 54-dimensional subspace of the 90 classes, different letters obstruct different subspaces, and together they obstruct all 90. Since the obstruction appears already at the first order of the lift, none of the 90 classes lies in the image of $\pi_{3 \rightarrow 2}$: every singlet BPS class at $(2, 2)$ is fortuitous.

This result is expected rather than discriminating, and the reason can be made precise. The projections preserve the degree, while the half-filling degree $k_* = 3pn^2/2$ moves by $3p(2n+1)/2$ when the rank grows by one, more than the width of any concentrated window. A rank- n class of degree $k_*(n)$ can only be monotone if there is a class of the same degree at rank $n+1$, so concentration at rank $n+1$ implies that every singlet BPS class at rank n is fortuitous; in this sense fortuity follows from concentration, much as in Chang–Chen–Sia–Yang’s argument for generic SYK [2]. At $(2, 2) \rightarrow (3, 2)$ the relevant rank-3 singlet space, at $k = 12$, has only 3494 states, and Section 7 checks the statement directly. The $(2, 2)$ test itself uses one coupling and floating-point ranks, and it is in the ordinary sense. Under the refined criterion of [14], classes of the form (determinant ground state) $\times$ (n -independent excitation) would count as generalised monotones; with fermionic letters many determinants vanish,

and whether any of the 90 classes is of that form has not been analysed.

5 The singlet spectrum: chaos at $p = 3$, hidden supersymmetry at $p = 2$

The exactness of the Casimir penalty makes the full singlet spectrum accessible at $n = 2$. Where the singlet subspace is small we construct it explicitly as $\ker P$ and diagonalise $(QU_k)^T(QU_k)$ on a singlet basis U_k , whose non-zero eigenvalues are the multiplet energies of the pair $(k, k+3)$. For the largest pair at $(2, 3)$, $(9, 12)$ with 10700 singlets in the lower degree, we project random vectors onto singlets with the exact polynomial $\prod_v \prod_{j=1}^6 (1 - J_v^2/j(j+1))$, which leaves a non-singlet residual of 1.4×10^{-15} , and compress $Q^T Q$ onto their span. Every count checks: the rank of the projected vectors equals $n(9)$, the kernel equals the rank of the previous map, and the lowest level, 151.34, agrees with Lanczos.

At $p = 3$ the multiplet spectra are chaotic. The mean ratio of consecutive level spacings over the central 80% of the 10024 multiplets of the pair $(9, 12)$ is $\langle r \rangle = 0.5268 \pm 0.0028$. The Gaussian-orthogonal value for matrices of the same size is 0.5343 ± 0.0026 and the Poisson value 0.3869 ± 0.0042 , so the spectrum is about 30 standard deviations from Poisson and within two of the orthogonal class. The smaller pair $(6, 9)$, with 676 levels, gives 0.507 ± 0.011 , and the locally unfolded spacing distribution follows the Wigner surmise (Figure 3c). The orthogonal class is the expected one, since for real couplings H is a real matrix in the occupation-number basis.

The near-BPS structure is equally clear. The lowest multiplet energies of the pairs at $(2, 3)$ are 0.05386 for $|q| = 1.5$, the pairs adjacent to the BPS degree, then 0.3348 for $|q| = 4.5$, and then a jump to the microscopic scale set by the couplings: 151.34, 575.23, 1506 and 2008 for $|q| = 7.5, 10.5, 13.5$ and 16.5. Small gaps exist, and only next to the BPS degree (Figure 2b). A quantitative test of the Turiaci–Witten density [4], $\rho_q(E) \propto \sinh(2\pi\sqrt{(E - E_0(q))/E_s})/E$ with $E_0(q) = q^2 E_s/4\hat{q}^2$, does not succeed at this size. Fixing $E_s = 0.862$ from the $|q| = 1.5$ edge predicts $E_0 = 0.485$ at $|q| = 4.5$, against 0.335 measured, an edge ratio of 6.2 instead of 9; fixing the normalisation from the eleven lowest $|q| = 1.5$ levels predicts 5.4, 258, 2000 and 8.7×10^4 multiplets with $|q| = 4.5$ below $E = 1.0, 2.17, 3.11$ and 5.38, against 2, 3, 5 and 7 measured (Figure 3d). This is not surprising. The Turiaci–Witten density is a statement at large e^{S_0} in which neighbouring charge sectors carry the same entropy, whereas at $(2, 3)$ the singlet sectors at $k = 12, 15$ and 18 hold 71685, 211113 and 300584 states, changing by factors of three to four per step. The test needs larger n . It also needs Gaussian couplings: the integer couplings used here have a non-zero mean, which takes the model out of the large- p SYK class (Section 8).

The case $p = 2$ behaves very differently, and the reason is an unexpected symmetry. The $(2, 2)$ spectra are full of exact coincidences between different pairs, and a search over cubic supercharges $\tilde{Q} = Q(\tilde{C})$ that commute with H on singlets finds exactly one besides Q , namely $\tilde{C} = (\varepsilon \otimes \varepsilon \otimes \varepsilon)\tilde{C}$ with ε the 2×2 antisymmetric tensor. On every singlet sector, to 10^{-15} , and for integer, Gaussian real and Gaussian complex couplings,

$$\tilde{Q}^2 = 0, \quad \{Q, \tilde{Q}\} = 0, \quad \{Q^\dagger, \tilde{Q}\} = 0, \quad \{\tilde{Q}, \tilde{Q}^\dagger\} = H, \quad (7)$$

which is an $\mathcal{N} = 4$ algebra. Off the singlets the last two relations fail by about 30% of $|H|$, so the extended supersymmetry is a property of the gauged model, closing only up to gauge transformations. The singlet spectrum organises completely into long $\mathcal{N} = 4$ multiplets spanning degrees $k, k+3, k+3$ and $k+6$, with $n(k) = b(k) + 2b(k-3) + b(k-6) + h(k)$ in every degree. Once the flavour symmetry $F = \varepsilon$ on every edge (or the corresponding antiunitary symmetry for complex couplings) is resolved, the level statistics are still not random-matrix like in any symmetry class, with coupling-independent exact degeneracies, and within the span of all gauge-invariant operators with at most four fermions the commutant of H on the $k = 9$ singlets is just $\{1, H\}$. At $p = 3$ the same search finds only Q . The hidden supersymmetry is plausibly tied to the pseudoreality of the $SU(2)$ flavour doublet, but we have not derived it; it holds at $n = 2$ and, on every singlet degree below half filling, at $n = 3$ (Section 7).

6 Large n : the singlet index as a three-species log gas

6.1 An exact rewriting

Write $x_{vi} = e^{i\theta_{vi}}$ for the eigenvalues of U_v in (4) at $t = -1$. Weyl’s integration formula introduces $|\Delta(x_v)|^2/n!$ for each node, and the edge factors become $\prod_{ij} (1 - e^{i(\theta_{vi} - \theta_{v+1,j})})^p$. For real φ , $1 - e^{i\varphi} = 2 \sin(\varphi/2) e^{i(\varphi-\pi)/2}$, and the φ -dependent phases cancel around the triangle, since each node is the source of one edge and the target

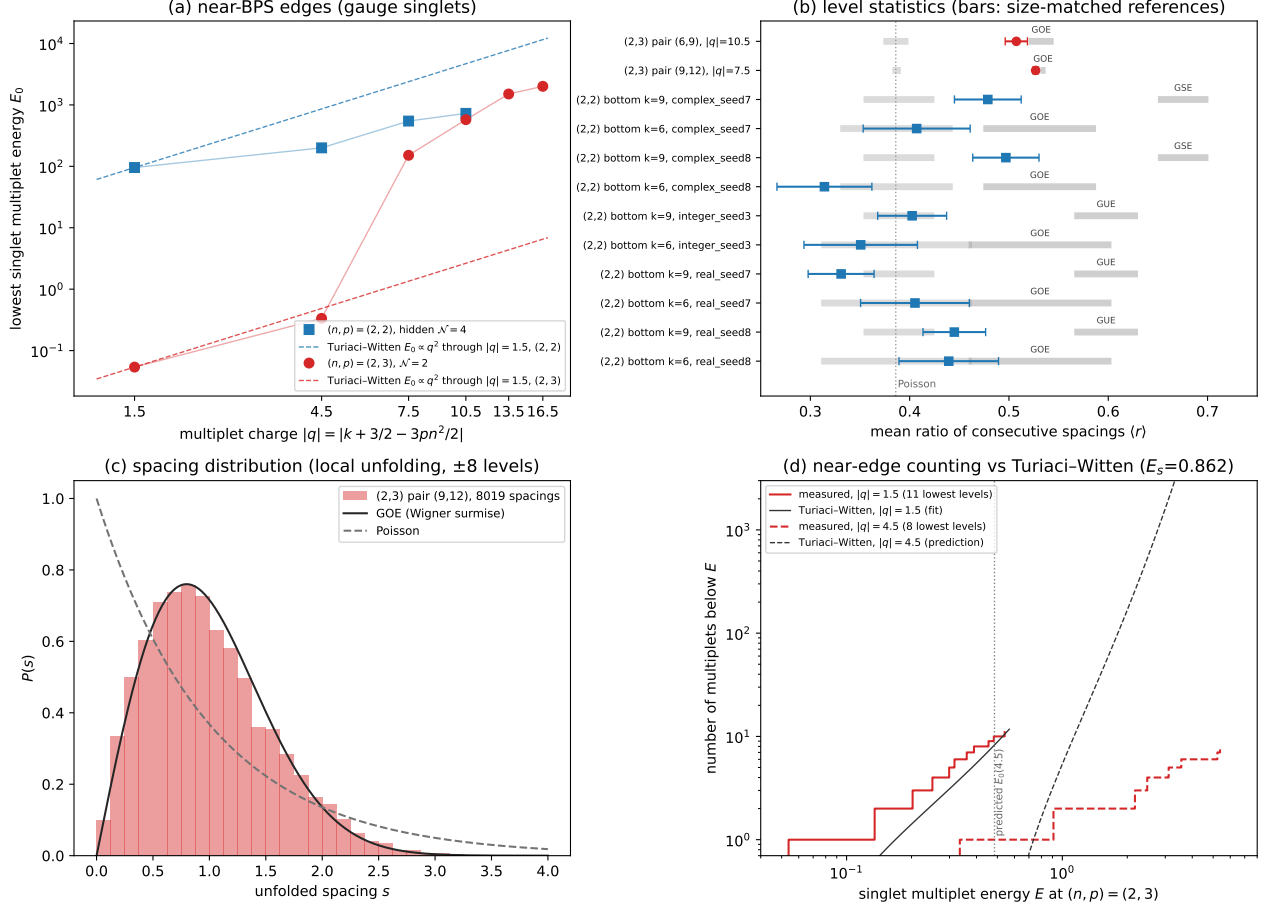


Figure 3: Singlet spectra at $n = 2$. (a) Lowest singlet multiplet energy of each Q -pair against its charge $|q|$, with the Turiaci–Witten q^2 law drawn through the $|q| = 1.5$ point. (b) Mean spacing ratios of every symmetry-resolved ensemble, with size-matched random-matrix and Poisson references. (c) Spacing distribution of the 10024 multiplets of the pair (9, 12) at (2, 3), locally unfolded, against the Wigner surmise and Poisson. (d) Counting functions of the two innermost multiplet sectors at (2, 3) against the Turiaci–Witten density, normalised on $|q| = 1.5$; the $|q| = 4.5$ curve is a prediction.

of another. Using $|x_a - x_b| = |2 \sin \frac{\theta_a - \theta_b}{2}|$,

$$I_0(n, p) = \frac{(-i)^{3pn^2}}{(n!)^3} \int \prod_{v,i} \frac{d\theta_{vi}}{2\pi} e^{L(\theta)} \sigma(\theta)^p, \quad (8)$$

$$L = \sum_{\text{pairs at the same node}} 2 \ln |2 \sin \frac{\theta_a - \theta_b}{2}| + p \sum_{\text{pairs at different nodes}} \ln |2 \sin \frac{\theta_a - \theta_b}{2}|,$$

with $\sigma = \prod_v \prod_{ij} \text{sgn} \sin \frac{\theta_{vi} - \theta_{v+1,j}}{2}$. This is the partition function of a gas of $3n$ charges of three species on the circle: charges of the same species repel with weight 2 and charges of different species with weight p , and for odd p there is an additional sign. Because every pair of distinct nodes is joined by exactly one edge, e^L is symmetric under permutations of the species, and only σ remembers the orientation of the quiver. The singlet count is the same integral with $|2 \cos \frac{\theta_a - \theta_b}{2}|$ for pairs at different nodes, which attract, and no phase.

Several exact statements follow immediately. At $p = 2$ every pair carries weight 2 and $\sigma^2 = 1$, so e^L is the Vandermonde $|\Delta_{3n}(x)|^2$ of all $3n$ eigenvalues together. Its integral is $(3n)!$, and

$$I_0(n, 2) = (-1)^n \frac{(3n)!}{(n!)^3} \quad \text{for every } n. \quad (9)$$

As measured by the index, the singlet BPS entropy of the $p = 2$ model is therefore $3n \ln 3 + O(\ln n)$, which is not

p	F^* (index)	arc half-width	G^* (singlets)	F^*/G^*	$F^*/3p$	$G^*/3p$
2	0 (exact)	$\pi/3$	1.4244	0	0	0.237
3	1.2373	1.0276	3.0427	0.41	0.137	0.338
4	2.5550	0.9717	4.7688	0.54	0.213	0.397
5	3.9297	0.9132	6.5623	0.60	0.262	0.437
6	5.3455	0.8606	8.4015	0.64	0.297	0.467
8	8.2619	0.7747	12.1714	0.68	0.344	0.507
16	20.4849	0.5796	27.8331	0.74	0.427	0.580
∞	$\frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4}$	$\sqrt{6/p}$	$3p \ln 2 - \frac{3}{2} \ln \frac{p}{2} - \frac{9}{4}$	$\frac{\ln 3}{2 \ln 2}$	$\frac{1}{2} \ln 3$	$\ln 2$

Table 2: Leading large- n coefficients, $\ln |I_0| = F^* n^2$ and $\ln(\#\text{singlets}) = G^* n^2$ up to $o(n^2)$, from the continuum solution, with the large- p asymptotics in the last row. The last two columns are per fermion mode.

macroscopic. The same manipulation, using $(1 - x\bar{y})^2 = -x\bar{y}|x - y|^2$ on the unit circle, gives for general p

$$I_0(n, p) = (-1)^n \frac{(3n)!}{(n!)^3} \mathbb{E}_{\text{CUE}(3n)} \left[\prod_v \prod_{i,j} (1 - x_{vi} \bar{x}_{v+1,j})^{p-2} \right], \quad (10)$$

where the $3n$ eigenvalues of a Haar-random $U(3n)$ matrix are split into three labelled groups. For even p the integrand of (8) is positive, so $I_0 \neq 0$ with sign $(-i)^{3pn^2} = (-1)^{pn^2/2}$; for odd p and odd n the prefactor is $\pm i$ while the integral is real, so $I_0 = 0$; and for odd p , Cauchy-Schwarz in p gives $|I_0(n, p)|^2 \leq |I_0(n, p-1)| |I_0(n, p+1)|$, which holds in the data with ratios 16.6 at (2, 3) and 1.62 at (2, 5). These are exactly the sign rules observed in Table 1.

6.2 The saddle point

At large n , with normalised eigenvalue densities ρ_v , $L = n^2 F[\rho] + O(n \ln n)$ with

$$F[\rho] = \sum_v \langle \rho_v, f \rho_v \rangle + p \sum_{v < w} \langle \rho_v, f \rho_w \rangle, \quad f(\alpha) = \ln |2 \sin \frac{\alpha}{2}| = - \sum_{m \geq 1} \frac{\cos m\alpha}{m}, \quad (11)$$

and since $(n!)^3 = e^{O(n \ln n)}$, $\ln |I_0| = n^2 \max F + o(n^2)$ for even p by standard log-gas large deviations. In terms of the moments $u_{v,m} = \int \rho_v e^{im\theta}$, $F = - \sum_m \frac{1}{m} u_m^\dagger [\mathbb{K} + \frac{p}{2}(J - \mathbb{K})] u_m$ with J the all-ones 3×3 matrix, whose eigenvalues are $1+p$ for the species-symmetric combination and $1-p/2$ twice. For $p < 2$ the uniform distribution is a stable maximum with $F = 0$; at $p = 2$ it is marginal; for $p > 2$ it is unstable towards configurations in which the species separate, in the spirit of the Hagedorn instability of weakly coupled gauge theories [19]. The maximum then sits on the boundary of the positivity constraint: the eigenvalues of the three nodes occupy three disjoint arcs rotated by $2\pi/3$ with respect to each other (Figure 4d). Physically the holonomies sit near $U_v \approx e^{2\pi i v/3}$, which turns the factor (-1) that each bifundamental fermion carries in the index into a phase $e^{\mp 2\pi i/3}$ and so evades the cancellations between even and odd states that keep the index small at $p \leq 2$.

We solve the continuum problem by expanding the density of one arc in Chebyshev polynomials. The Cauchy part of the Euler-Lagrange equation is treated exactly and the smooth part by quadrature; the arc width follows from the condition that the edges be soft, and F^* is three times the Lagrange multiplier. The solution converges to twelve digits, and the effective potential is flat on the support to 10^{-15} and lower in the gaps. Independently, Newton maximisation of the discrete L up to $n = 2048$ extrapolates to the same coefficients to 10^{-9} or better, and for $n \leq 12$ no unconstrained random start finds a higher maximum than the symmetric one. The singlet count is treated in the same way: there the three species coincide on a single arc. Table 2 gives the coefficients.

Both limits can be derived analytically and are reproduced by the numerics. Near $p = 2$ the uniform arcs of length $2\pi/3$ give the bound $F^*(p) \geq (p/2 - 1) 39\zeta(3)/2\pi^2$, which is tight as $p \rightarrow 2^+$. At large p the arcs shrink to width $O(p^{-1/2})$, each becomes a semicircle of radius $\sqrt{6/p}$ in the effective harmonic potential of the other two, and $F^* = \frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4} + O(1/p)$; the singlet count behaves in the same way with $\ln 2$ in place of $\frac{1}{2} \ln 3$. The leading terms have a simple meaning. $3p \ln 2 n^2$ is the logarithm of the dimension of the whole Fock space, and $\frac{3p}{2} \ln 3 n^2$ is what $3pn^2$ fermion modes contribute when each carries the weight $|1 - e^{-2\pi i/3}| = \sqrt{3}$. The BPS entropy per fermion therefore approaches $\ln(2 \cos \frac{\pi}{6}) = \frac{1}{2} \ln 3$, which is precisely the ground-state entropy per complex fermion of $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ [3], and the ratio F^*/G^* approaches the SYK ratio $\ln \sqrt{3}/\ln 2 = 0.79$. The same constant appears in Tierz's large- N result for the irrep-summed indices of the single-matrix $\text{Tr} \Psi^3$ model

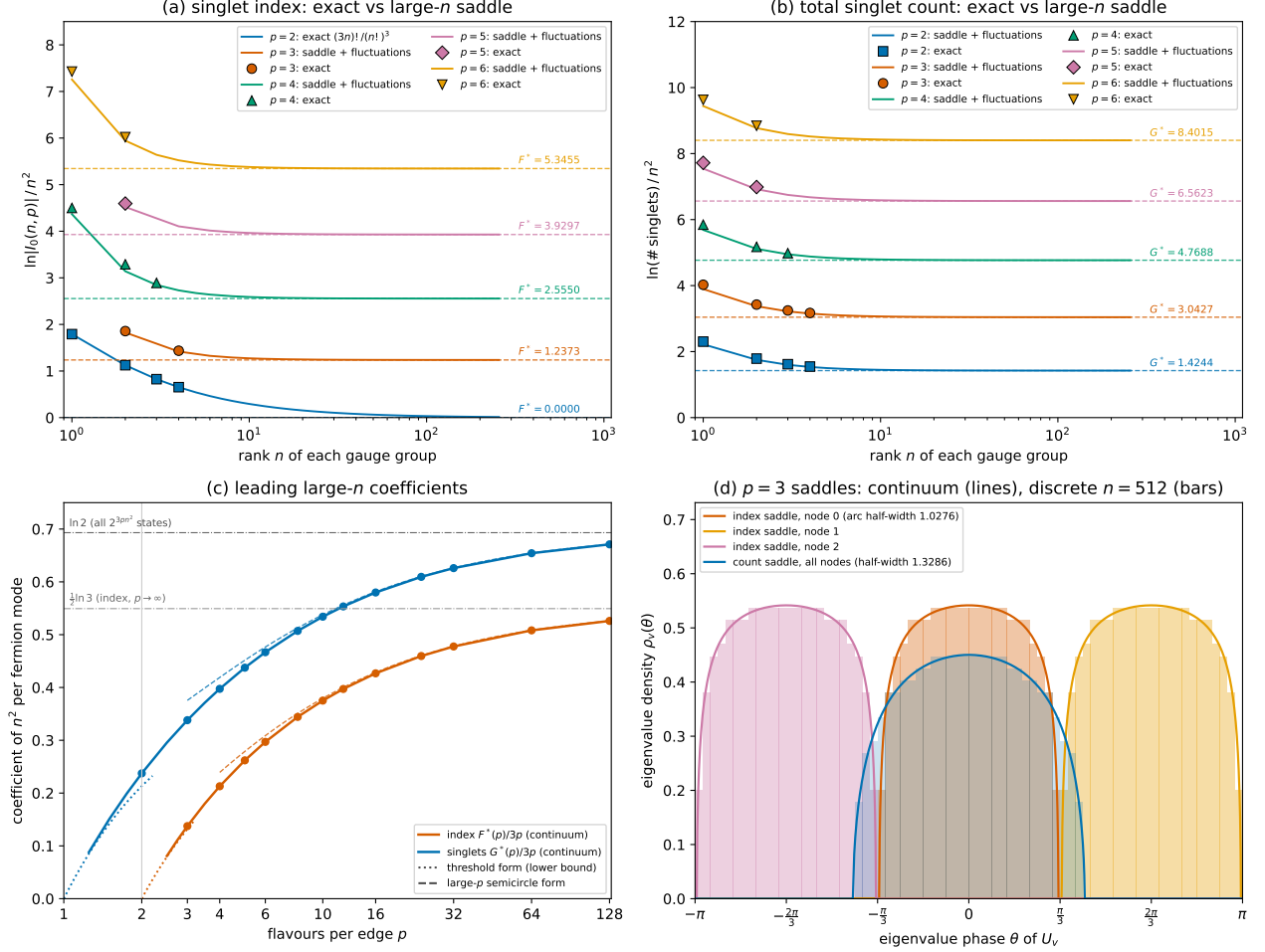


Figure 4: The large- n saddle. (a) $\ln|I_0|/n^2$ against n : exact values (markers), the saddle-point estimate with Gaussian fluctuations and the universal crystal correction (lines), and the asymptotes $F^*(p)$ (dashed); for $p=2$ the line is the exact result (9). (b) The same for the total singlet count. (c) The leading coefficients per fermion mode against p , with the threshold bound and the large- p asymptotics. (d) The equilibrium densities at $p=3$: the three index arcs and the single arc of the count saddle, with histograms of the discrete maximum at $n=512$.

[13]. At finite p the gauge constraint costs $\frac{1}{2p} \ln \frac{2p}{3} + \frac{3}{4p}$ per mode, which is large at $p=3$: there the singlet BPS entropy per mode is 0.137, a quarter of the SYK value, but it is a finite fraction, 41%, of the singlet log-count.

6.3 Finite n and the test at $(4, 3)$

The coefficient F^* alone cannot be compared with the data at $n \leq 4$: at $(4, 3)$ it gives $F^*n^2 = 19.80$ against the exact $\ln|I_0| = 22.97$, a factor of 24 too small. The subleading terms come from the same saddle. The discrete maximum behaves as $L_{\max} = F^*n^2 + 3n \ln n + 3n \int g \ln(2\pi g) + \frac{1}{4} \ln n + O(1)$, where g is the density of one arc: the $n \ln n$ and $O(n)$ terms are the self-energy of each species at local spacing $1/(ng)$, which we verified to six digits, and the $\frac{1}{4}$ is consistent with $\frac{1}{24}$ per soft edge. Adding the Gaussian fluctuations about the maximum overshoots by a universal amount, the excess of the harmonic approximation for a $\beta=2$ log gas, which on the circle is $\ln(e/\sqrt{2\pi})$ per eigenvalue up to $O(1/n)$. With that correction the estimate reproduces (9) to $1/36n$, and for $p \neq 2$ we assume that each arc is locally a $\beta=2$ gas, which is the one heuristic step. The resulting estimate has no free parameters, and Table 3 compares it with every non-zero exact value we have. All 27 residuals are negative and none exceeds 0.6 in the logarithm, across values from $e^{1.8}$ to $e^{50.7}$; at $(4, 3)$ the estimate gives 22.69, within 25% of $|I_0| = 9.43 \times 10^9$. At large n the $O(n)$ terms cancel to fit precision and the estimate takes the form $\ln|I_0| \approx F^*n^2 + \frac{1}{2} \ln n + d(p)$, with $d \approx 2.2$ – 2.4 for $p=3$ – 6 , in line with the absence of an $O(n)$ term in $\beta=2$ ensembles. For $p=3$ it predicts $\ln|I_0| \approx 47.7, 82.6, 127.3$ and 181.8 at $n=6, 8, 10$ and 12 , beyond the reach of the exact recursion.

(n, p)	$\ln I_0 $			$\ln(\#\text{singlets})$		
	exact	saddle	difference	exact	saddle	difference
(1, 2)	1.79	1.76	-0.03	2.30	2.22	-0.08
(2, 2)	4.50	4.49	-0.01	7.15	7.00	-0.15
(3, 2)	7.43	7.42	-0.01	14.56	14.40	-0.17
(4, 2)	10.45	10.45	-0.01	24.75	24.55	-0.19
(1, 3)		both vanish		4.03	3.90	-0.13
(2, 3)	7.43	7.30	-0.12	13.70	13.50	-0.20
(3, 3)		both vanish		29.22	28.97	-0.25
(4, 3)	22.97	22.69	-0.28	50.74	50.44	-0.30
(1, 4)	4.50	4.37	-0.13	5.85	5.69	-0.16
(2, 4)	13.16	12.56	-0.60	20.69	20.45	-0.24
(3, 4)	26.02	25.62	-0.39	44.85	44.54	-0.30
(1, 5)		both vanish		7.72	7.54	-0.18
(2, 5)	18.38	18.08	-0.29	27.94	27.68	-0.26
(1, 6)	7.43	7.26	-0.17	9.63	9.44	-0.19
(2, 6)	24.08	23.78	-0.30	35.37	35.09	-0.28

Table 3: The saddle-point estimate (Gaussian fluctuations about the discrete saddle, minus the universal $\beta = 2$ crystal excess, no free parameters) against all exact values. For odd p and odd n the index and the estimate both vanish exactly, the latter because the two orientations of the arcs cancel.

The status of these statements should be clear. The rewriting (8), the formula (9) and the sign rules are exact. The coefficient F^* for even p rests on standard large-deviation results for log gases, which we have not reproved. For odd p it also requires that the sign σ^p does not cancel the leading saddle. Sign changes need an eigenvalue of one node to pass an eigenvalue of a neighbouring node across the gap between arcs, which costs an action of order n ; but the gap at $p = 3$ is only 0.039 radians, so the suppression is modest at the sizes of interest. The exact values at $(2, 3)$ and $(4, 3)$ show no sign of cancellation, their residuals being like those at even p . Finally, $|I_0|$ is a lower bound on the number of singlet BPS states, equal to it only in the absence of cancellations between degrees, for example under concentration; and for odd pn^2 the index vanishes and says nothing about that number.

7 Beyond $n = 2$: a Schur–Weyl basis and a singlet bootstrap

7.1 Exact methods at rank three

Two tools take us past $n = 2$. The first is a singlet-adapted basis. By Schur–Weyl duality a singlet of degree $3m$ is a triple of elements of the group algebra of S_m , one for each node, recording how the m letters leaving the node are contracted with the m letters arriving at it. In Young’s orthogonal form the fermionic relabellings become sign projections on each edge, the finite- n trace relations become the condition that every Young diagram has at most n rows, and Q becomes a Kronecker product of Gelfand–Tsetlin inclusions, which is applied without ever building Fock vectors. The basis reproduces every singlet count and every stored $n = 2$ spectrum. At $(3, 2)$ Lanczos diagonalisation in this basis finds no singlet BPS states in any degree $k \leq 24$, so all $|I_0| = 1680$ of them sit at half filling, $k = 27$: in the $p = 2$ corner concentration survives the step to the next rank. Since the rank-3 cohomology vanishes at $k = 12$, the argument of Section 4.3 makes all 90 classes at $(2, 2)$ fortuitous to all orders, and the hidden $\mathcal{N} = 4$ of Section 5 holds at $n = 3$ on every singlet degree below half filling. At $p = 3$, however, the basis reaches $(3, 3)$ only for $k \leq 12$, where it again finds no BPS singlets (Table 5), because the singlet dimension grows to 1.2×10^8 at $k = 18$ and 10^{12} at half filling. Nor is there an algebraic shortcut. Upper semicontinuity bounds the generic cohomology by that of any special coupling, but the natural special coupling $C_{abc} = \delta_{ab}\delta_{bc}$, for which the singlet complex is the gauge-invariant part of a tensor power of the $p = 1$ complex, is not concentrated even at $(2, 3)$, where its singlet BPS states spread over the degrees $12 \leq k \leq 24$.

7.2 The semidefinite program

The second tool is a semidefinite bootstrap of the singlet sector, built on the trace-word algebra of [7]. Write $\Psi_{e,f}$ for the matrix of creation operators of edge e (the fields A, B, C for $e = 0, 1, 2$) and flavour f , whose row index lives at node e and column index at node $e + 1$, and $\bar{\Psi}_{e,f}$ for the matrix of adjoints, $(\bar{\Psi}_{e,f})_{yx} = (\Psi_{e,f})_{xy}^\dagger$. The only non-vanishing anticommutator contracts two indices of the same node, so products of these matrices are brought

to a canonical form exactly as for the adjoint models, with a factor n for every closed index loop. Gauge-invariant operators are products of traces of closed walks on the triangle, and for a density matrix ρ supported on singlets of degree $k = 3m$ the expectation values $\phi(X) = \text{Tr} \rho X$ of these products are the unknowns. With real couplings every such operator is a real matrix in the occupation basis and every eigenspace of H has a real basis, so ϕ may be taken real, which halves the problem relative to the complex engine of [7].

Positivity enters through Gram matrices. For open words W_a , products of letters that form a walk from node u to node v , and for traced closed walks T_a together with the identity,

$$M_{ab} = \phi(\text{Tr} W_a^\dagger W_b) - \frac{\delta_{uv}}{n} \phi(\text{Tr} W_a^\dagger \text{Tr} W_b) \succeq 0, \quad S_{ab} = \phi(T_a^\dagger T_b) \succeq 0. \quad (12)$$

M is the Gram matrix of the vectors $(W_a)_{ij} \psi$ summed over the open indices, with the trace part removed when the word returns to its starting node. Words with different endpoints cannot be combined, and words with different edge charges have orthogonal images because ρ has definite N_A , N_B and N_C , so both matrices are block diagonal in these labels. The linear constraints are the normalisation $\phi(1) = 1$; reality; the sector conditions $\phi((N_e - m)X) = 0$ for the three edge numbers; the gauge conditions $\phi(\hat{C}_v X) = 0$, where $\hat{C}_v = n \text{Tr} X_v^2 - (\text{Tr} X_v)^2$ is n times the $SU(n)$ Casimir of node v , built from its generator matrix $X_v = \sum_f (\Psi_{v,f} \bar{\Psi}_{v,f} + \bar{\Psi}_{v-1,f} \Psi_{v-1,f})$; and the equations of motion $\phi([H, X]) = 0$, which hold for any mixture of eigenstates. Minimising $\phi(H)$ subject to all of these gives a lower bound on the lowest singlet energy $E_0(k)$. For exclusion we add the BPS conditions

$$\phi(XQ) = \phi(QX) = \phi(XQ^\dagger) = \phi(Q^\dagger X) = \phi(XH) = \phi(HX) = 0, \quad (13)$$

which every BPS state satisfies. A singlet BPS state in degree k would then supply a feasible ϕ , so infeasibility proves $h_{\text{sing}}^k = 0$. We test it in two ways: through the margin t^* , the largest t for which every Gram matrix can be made $\succeq t \mathbf{1}$, with $t^* < 0$ meaning infeasible; and through the pure feasibility problem, whose infeasibility comes with a Farkas certificate. Writing the problem as $Ax + s = b$ with s in the product K of a zero cone and the semidefinite cones, the certificate is a vector y with $A^T y = 0$, $y \in K^*$ and $b \cdot y < 0$. Exclusion at one coupling suffices: $h^k = \dim V_k - \text{rk } Q_k - \text{rk } Q_{k-3}$ is upper semicontinuous in C , so its vanishing at our integer couplings implies its vanishing for generic couplings.

All results below are at what we call level 2: open words of length at most two, closed walks of length at most three, and constraints generated by operators of length at most four. At $p = 3$ the largest Gram blocks then have 37, 27 and 18 rows, and the exclusion problem has 1.6×10^5 unknowns and 1.2×10^5 equality constraints after the reality reduction. The interior-point solver Clarabel solves it in one to nine minutes at a peak of about 3 GB. Every family of constraints has been evaluated on exact functionals, computed from Fock-space states at $(n, p) = (1, 2)$, $(1, 3)$ and $(2, 1)$, and holds to 10^{-15} .

7.3 Validation at $n = 2$

Table 4 compares the bootstrap with the exact singlet spectra at $n = 2$. There are no contradictions: no degree that carries BPS states is excluded, and every energy bound lies below the exact gap, by 17% and 25% at $k = 3$ and by more towards the frontier. Level 2 excludes $k \leq 6$ at $(2, 2)$ and $k \leq 9$ at $(2, 3)$, in both cases half of the way to half filling. At $(2, 3)$ it stops precisely where the exact gap collapses from 151 to 0.33, but at $(2, 2)$ it also stops at $k = 9$, where the gap is still 95.6; we return to this below.

7.4 Results at $(3, 3)$

At $(3, 3)$ the problem is the same as at $(2, 3)$ apart from the value of n in its coefficients, and Table 5 and Figure 5 give the results. Where exact values exist, $k \leq 12$, the energy bounds again lie below them. The margin falls almost linearly in k , in steps that shrink from 0.98 to 0.85, and crosses zero between $k = 21$ and 24. The solves at $k \geq 24$, including the degrees next to half filling where BPS states may exist, are all feasible, as they must be if the method is sound. In the two decisive degrees, $k = 18$ and 21, the pure feasibility problem returns Farkas certificates whose dual semidefinite blocks are positive semidefinite, and the energy bounds there are positive as well.

The certificates are robust in a precise sense. If a singlet BPS state existed, its functional x^* would satisfy $Ax^* + s = b$ with $s \in K$, so that $0 \leq \langle y, s \rangle = b \cdot y - (A^T y) \cdot x^*$. Every monomial of total length L obeys $|\phi(m)| \leq \|m\| \leq n^L$, with $L \leq 8$ here, so $\|x^*\| \leq 3^8 \sqrt{1.6 \times 10^5} \approx 2.6 \times 10^6$, and the right-hand side is at most $(-4.0 \times 10^{-3} + 6 \times 10^{-5})\|y\| < 0$ at $k = 18$. This is a contradiction with a factor of about 70 to spare, and the same holds at $k = 21$. We have since repeated the check in exact integer arithmetic. The constraint coefficients

(n, p)	k	exact E_0	bound on E_0	margin t^*	verdict
(2, 2)	3	546	455.1	-1.334	excluded
	6	199.95	182.6	-0.508	excluded
	9	95.56	0	0	not excluded
	12	0 (90 BPS states)	0	0	BPS degree, kept
(2, 3)	3	1506	1129.3	-1.342	excluded
	6	575.23	253.7	-0.732	excluded
	9	151.34	14.4	-0.163	excluded
	12	0.3348	0	0	not excluded
	15	0.0539	0	0	not excluded
	18	0 (1680 BPS states)	0	0	BPS degree, kept

Table 4: Validation of the level-2 singlet bootstrap against the exact singlet spectra at $n = 2$, for the seed-3 integer couplings used throughout. The bound is the minimum of $\phi(H)$ without the BPS conditions; the margin is computed with them, and $t^* = 0$ stands for $|t^*| < 10^{-6}$. Degrees above half filling follow by particle-hole symmetry.

k	$n(k)$	exact E_0	bound on E_0	t^*	$\ A^T y\ /\ y\ $	$b \cdot y/\ y\ $	verdict
3	27	6024	5457.3	-5.894			excluded
6	702	4634.9	4139.9	-4.912			excluded
9	18 981	3546.0	2824.4	-3.956			excluded
12	460 539	2472.0	1511.2	-3.012			excluded
15	8.7×10^6	—	598.2	-2.102			excluded
18	1.2×10^8	—	218.6	-1.219	2.3×10^{-11}	-4.0×10^{-3}	excluded
21	1.2×10^9	—	21.8	-0.374	6.1×10^{-12}	-1.2×10^{-3}	excluded
24	8.4×10^9	—	0	0	no certificate		not excluded
27–39	up to 1.1×10^{12}	—	0	0			not excluded

Table 5: The level-2 singlet bootstrap at $(3, 3)$, seed-3 couplings, $E_{\text{vac}} = 6777$. Exact energies for $k \leq 12$ come from the Schur–Weyl basis. The two certificate columns characterise the Farkas vectors of the pure feasibility problem. The energy bound at $k = 21$ was obtained at reduced solver accuracy. By particle-hole symmetry the exclusions extend to $k \geq 60$.

are rationals whose denominators divide $3 \cdot 2^{20}$, the certificate is rounded to a dyadic grid, and each dual block is shifted by at most 10^{-12} of its size until it is positive definite, which is verified exactly by Sylvester’s criterion. The bound $\sum_j |r_j| n^{L_j}$ then comes to 1.0×10^{-7} and 2.4×10^{-7} of $|\mu \cdot b|$ at $k = 18$ and 21 , and an independent certificate from the first-order solver SCS verifies as well. The exclusions are therefore proofs, conditional only on the correctness of the constraint generation, which is tested against explicit Fock-space operators.

Two conclusions follow. The first is fortuity at $p = 3$. The singlet cohomology of the $(3, 3)$ model vanishes in degree 18, which is half filling at $(2, 3)$, so by the argument of Section 4.3 none of the 1680 singlet BPS classes of the $(2, 3)$ model has a rank-3 preimage. All of them are fortuitous in the ordinary sense, with the caveat of the refined criterion of [14]. The second is partial concentration at rank three. There are no singlet BPS states at $(3, 3)$ for $k \leq 21$ and, by particle-hole symmetry, none for $k \geq 60$. The degrees $24 \leq k \leq 36$ and their mirror images remain open, besides the middle pair $k = 39, 42$, which carries equal numbers of BPS states because $I_0(3, 3) = 0$, possibly none.

7.5 Where the bootstrap stops

Level 2 stops about half-way to half filling at all three points, much as it did for the adjoint models [7], and $(2, 2)$ shows why. Diagonalising H on the full Fock sector $N_A = N_B = N_C = m$, with every gauge representation included, we find no BPS states at all for $m = 2$, whose lowest level is 70.9, but at least twelve for $m = 3$, all of them gauge non-singlets, since the singlet gap at $k = 9$ is 95.6. The frontier thus sits exactly at the onset of non-singlet BPS states in the sector. This is natural, because positivity alone cannot tell a singlet from the gauge average of a non-singlet: the matrix M in (12) is positive for any state, and gauge invariance enters only through the Casimir conditions. Accordingly those conditions barely affect the exclusions. Switching them off changes the margins at $k = 3$ and 6 from -1.334 and -0.508 to -1.327 and -0.505 , although it lowers the energy bounds from 455 and 183 to 293 and 63, so the exclusions obtained so far rule out BPS states in every gauge representation at once. None of the extensions we have tried moves the $(2, 2)$ frontier: levels 3 and 4, with Gram blocks of 26 and 104 words; Gauss-law conditions $\phi(Y \text{Tr}[W G_v]) = 0$, with G_v the $u(n)_v$ generator matrix and

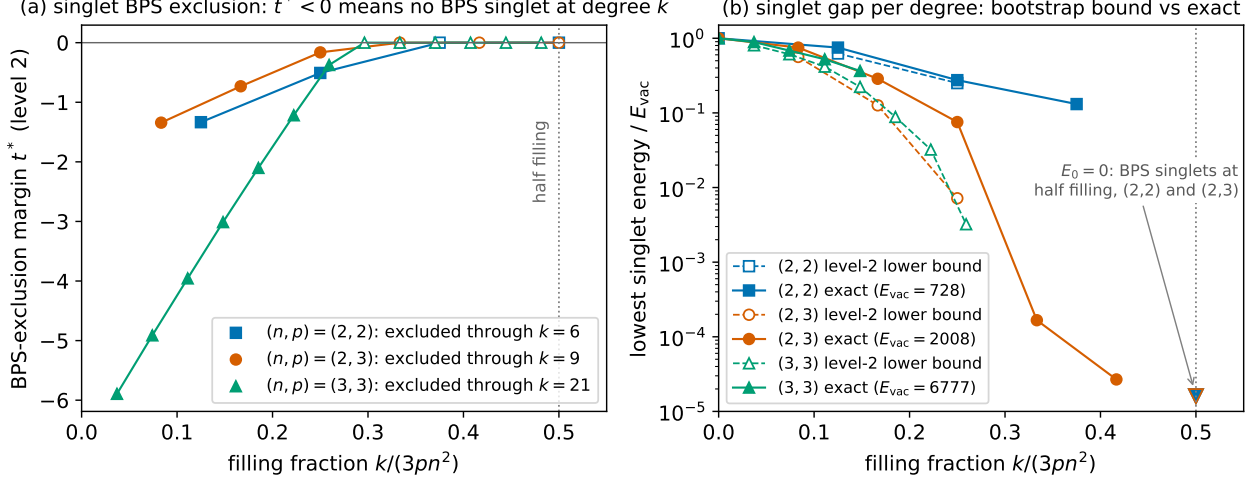


Figure 5: The singlet bootstrap at level 2. (a) The BPS-exclusion margin t^* against the filling fraction $k/3pn^2$ at $(2,2)$, $(2,3)$ and $(3,3)$; filled markers are excluded degrees ($t^* < 0$) and open markers are not. (b) The lowest singlet energy per degree in units of the vacuum energy $E_{\text{vac}} = n^3 \sum_{abc} C_{abc}^2$: bootstrap lower bounds (dashed, open markers, with vanishing bounds omitted) against the exact values (solid) where these are known; the triangles at half filling mark the BPS degrees at $n = 2$.

n	half filling	excluded through k (filling)	not excluded at k (filling)	exact slack $\sum_j r_j n^{L_j} / \mu \cdot b $
2	18	9 (0.250)	12 (0.333)	4.5×10^{-8}
3	40.5	21 (0.259)	24 (0.296)	2.4×10^{-7}
4	72	39 (0.271)	42 (0.292)	2.0×10^{-6}
5	112.5	60 (0.267)	63 (0.280)	2.7×10^{-6}
6	162	87 (0.269)	90 (0.278)	1.1×10^{-5}

Table 6: The level-2 BPS-exclusion frontier at $p = 3$, seed-3 couplings. The last column is the slack of the exact verification of the certificate at the last excluded degree; any value below 1 proves the exclusion.

W an open word from v back to v ; the finite- n trace relations; and the statement that for a singlet ψ each vector $W_{ij}\psi$ lies in a definite gauge representation, so that $\sum_{ij} \langle W_a \psi | \hat{C}_x | W_b \psi \rangle$ is a fixed multiple of M_{ab} . The last of these strengthens the margin at $k = 6$ by half but leaves $k = 9$ feasible, and at $(3,3)$ it likewise leaves $k = 24$ feasible. That level 4 fails as well settles the question in this formulation: going further needs constraints that tie positivity to the gauge representation of the state much more tightly than Casimir conditions do, not more operators, and concentration near half filling must come from other tools.

7.6 The frontier at larger rank

The level-2 problem has the same size at every rank; only its coefficients depend on n . At $n \geq 4$ the interior-point solver needs rescaled variables, $x_j = n^{w_j} y_j$ with $w_j = \sum (1 + L/2)$ the 't Hooft weight of the monomial, because the monomial values span about n^8 . With them each feasibility solve takes about a minute, and column scaling changes neither the feasible set nor the Farkas vector. Locating the frontier by bisection and verifying the certificate at the last excluded degree exactly gives Table 6. The excluded filling fraction settles near 0.27, so at every rank we have tested the singlet BPS states, if there are any, are confined to the central 46% of the filling range. The convergence is an extrapolation from five ranks, but each entry is a certificate.

Two further rank-fortuity statements follow. The $(4,3)$ model has no singlet BPS states at $k = 39$, so any BPS classes of the $(3,3)$ model at $k = 39$ are fortuitous; their partners at $k = 42$ lie beyond the frontier at $(4,3)$. The $(3,4)$ model has none at $k = 24$ (verified exactly, for a problem of 1.8×10^6 monomials at a peak memory of 9.7 GB), so every singlet BPS class of the $(2,4)$ model at half filling is fortuitous. Since the half-filling degree of rank n sits at filling $n^2/2(n+1)^2$ of rank $n+1$, a fraction that tends to $1/2$, level 2 can prove fortuity of half-filling classes only for the first steps in n .

p	k	$\dim H^k(p)$	horizontal: $\dim H^k(p+1)$, rank π_*	diagonal: $\dim H^{k+3}(p+1)$, rank ι_*	through neither
2	3	6	2, 2	2, 2	2
3	3	2	0, 0	90, 2	0
3	6	2	90, 2	0, 0	0
4	6	90	0, 0	0, 0	90

Table 7: Uplift channels in the arrow direction at $n = 1$, nested Gaussian couplings. The last column is the dimension of the classes that uplift through neither channel. $p = 5$ has no BPS states; the 1680 classes at $p = 6$ need the central cohomology of $p = 7$, whose blocks of 42875 states have a nearly gapless lowest multiplet and are not yet resolved.

7.7 Fortuity in the rank and arrow directions

Which family defines fortuity deserves a definite answer, because the companion study of supersymmetric SYK [18] tests it differently. There the system grows by adding fermions at fixed charge. In its discussion of the Abelian three-node quiver with m arrows per edge, the rank is fixed at one and m grows at fixed cohomological degree. Ambient classes then persist, while the variable, pure-Higgs classes occur only at a single value of m , a fortuity window of width one, for which a diagonal uplift is proposed. Here we fix p and grow the rank. Both are legitimate, since fortuity is defined relative to a family with degree-preserving cochain maps, and the projection that deletes a new flavour is such a map for nested couplings, just as Tierz’s projection is for a new index. But the two families have different limits, the planar limit and the SYK-like large- p limit, and their verdicts can differ for the same class. What distinguishes them in practice is how fast the half-filling degree $3pn^2/2$ moves compared with the singlet spacing of 3: by $3p(2n+1)/2 \geq 9$ per unit of rank, but by $3n^2/2$ per added flavour. Under concentration a rank step, or a flavour step at $n \geq 2$, leaves no class behind. A flavour step at $n = 1$ moves the window by only half a degree, which is the slow regime of [18], where walks and diagonal uplifts make sense.

Our $n = 1$ member is precisely the fermionic version of the quiver example of [18], with the same Abelian cyclic quiver and superpotential but fermionic matter. Its singlet BPS states sit in the central blocks only: 6 at $p = 2$, 2 + 2 at $p = 3$, 90 at $p = 4$, none at $p = 5$, 1680 at $p = 6$. There is therefore no ambient, monotonous row; in the fermionic model fortuity covers the whole singlet BPS sector, as in generic $\mathcal{N} = 2$ SYK. Adding flavour p gives the counterparts of the two uplift channels of [18]. The horizontal channel deletes the new flavour, $\pi_* : H^k(p+1) \rightarrow H^k(p)$. The diagonal channel multiplies by $a_p^\dagger b_p^\dagger c_p^\dagger$; the fully occupied states form a sub-complex, and $Q\iota = -\iota Q$ exactly, so this gives $\iota_* : H^k(p) \rightarrow H^{k+3}(p+1)$. In SYK every class uplifts through at least one channel, which here is the statement $\ker \iota_* \subseteq \text{im } \pi_*$. Table 7 gives the exact ranks for nested Gaussian couplings. At $p = 3$ the SYK pattern holds: the lower central class lifts diagonally and the upper one horizontally. It fails as soon as the smaller theory has even p : 2 of the 6 classes at $p = 2$, and all 90 at $p = 4$, uplift through neither channel. The reason is that a flavour adds three modes coupled by the singlet constraint, so the partially occupied pieces of the filtration by new-flavour occupation carry non-singlet remainders, and the two extreme channels no longer exhaust the long exact sequence of the SYK case. The fermionic quiver at $n = 1$ is thus unlike both the bosonic quiver and SYK.

We therefore adopt the rank direction as the definition of fortuity for the claims of this report, which concern the planar limit, and every fortuity statement in Section 4.3 and in this section is rank fortuity at fixed p . The arrow direction is the notion of [18]. At $n \geq 2$ it adds nothing beyond concentration: the 90 classes at (2, 2) are arrow-fortuitous as well, since (2, 3) has no singlet cohomology at $k = 12$. At $n = 1$ it is genuinely different, and Table 7 is its exact form.

8 Large p : the Schwinger–Dyson saddle and the full $n = 1$ model

At fixed n the flavour indices of the quiver play the role of the site indices of SYK, and the limit of large p is controlled by a saddle point. Take the couplings independent and Gaussian with zero mean and $\langle C_{abc}^2 \rangle = J/(p^2 n)$. Averaging over them and introducing flavour-averaged bilocal fields that are matrices in colour, $G_A(\tau, \tau')_{ij, i'j'} = \frac{1}{p} \sum_a \bar{A}_{ij}^a(\tau) A_{i'j'}^a(\tau')$ and likewise for B and C , turns the action into p times a functional of the bilocals. At the colour-symmetric saddle every colour loop contributes a factor n , and each species obeys $\Sigma_A = JG_B G_C$ and its cyclic images. These are the Schwinger–Dyson equations of $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ [3], for

$$N = 3pn^2 \text{ complex fermions}, \quad J = p^2 n \langle C^2 \rangle. \quad (14)$$

The singlet projection adds an integral over the three holonomies, of dimension $3n^2$ against an action of order pn^2 , so it changes the free energy only at relative order $1/p$ up to logarithms, as it does for the index in Section 6.

The normalisation is fixed exactly at infinite temperature. With $O_{abc} = \text{Tr}(A^a B^b C^c)$ one has $\langle O_{abc} O_{a'b'c'}^\dagger \rangle_\infty = \delta_{aa'} \delta_{bb'} \delta_{cc'} n^3/8$, hence $\langle H \rangle_\infty/N = J/12$ for every coupling draw and every p and n . The argument is the standard one for SYK. It is not rigorous here, because each coupling is shared by n^3 colour triples, and we have not checked the dominance of melonic diagrams diagram by diagram. In this section we solve the saddle, derive its predictions, and test them against the $n = 1$ model, which we diagonalise exactly at $p \leq 6$ in all of its gauge-charge sectors.

8.1 The saddle and its predictions

At zero chemical potential for the R-charge the three species are equal, and the complex equations reduce to the $\mathcal{N} = 1$ equations of [3],

$$\Sigma_\psi = (\hat{q} - 1)J \tilde{G}_b G_\psi^{\hat{q}-2}, \quad \Sigma_b = J G_\psi^{\hat{q}-1}, \quad G_\psi(i\omega) = \frac{1}{-i\omega - \Sigma_\psi(i\omega)}, \quad G_b(i\nu) = -\frac{1}{1 + \Sigma_b(i\nu)}, \quad (15)$$

with $G_b = -\delta + \tilde{G}_b$, while the action per complex fermion is twice the $\mathcal{N} = 1$ action per Majorana fermion. The energy needs one remark. Since $H = Q^2$ and Q is linear in the couplings, $J\partial_J \log Z = -\beta\langle H \rangle$, and per Majorana fermion

$$\frac{E}{N} = \frac{J}{\hat{q}2^{\hat{q}}} - \frac{J}{2} \int_0^\beta \tilde{G}_b(\tau) G_\psi(\tau)^{\hat{q}-1} d\tau. \quad (16)$$

The integral comes from the smooth part of G_b . The delta-function part multiplies an ordering-ambiguous product of propagators at coincident points, and it is fixed by the operator statement that the c -number part of Q^2 is $\sum_I C_I^2/2^{\hat{q}}$, which is the first term. The slope formula (2.37) of [3] omits this constant; without it the energy of the supersymmetric ground state does not vanish.

We solve (15) on a midpoint grid in imaginary time with fast Fourier transforms, subtracting the free tail analytically, on grids with $\beta J/M \leq 0.002$, with Richardson extrapolation in M and a spline quadrature for the thermodynamic integrals. Three checks fix the accuracy. The zero-temperature entropy $\frac{1}{2} \ln 2 - \int_0^\infty E d\beta$ comes out as 0.2746523 per Majorana fermion against the exact $\frac{1}{2} \ln(2 \cos \frac{\pi}{6}) = 0.2746531$ (Fu et al. found 0.2745 ± 0.0005 numerically), and as 0.3214821 against 0.3214827 at $\hat{q} = 5$. At high temperature the solution reproduces $E \rightarrow J/24$ and $dE/d\beta \rightarrow -J^2/32$, the exact moments derived below. At $\beta J = 100$ the propagator at $\tau = \beta/2$ is within 0.3% of the conformal solution. The low-temperature energy then supplies the one number that the $\mathcal{N} = 2$ super-Schwarzian leaves free [3]. A fit $E = \sum_{k=2}^5 a_k T^k$ on $\beta J \geq 50$ gives $a_2 = 0.0833381$ per Majorana fermion, and in the convention $E/N \rightarrow 2\pi^2 \alpha_s T^2/J$

$$\alpha_s = 0.0084439 \quad \text{per complex fermion}, \quad (17)$$

stable to 10^{-4} across fit windows. Fu et al. write down the Schwarzian action but do not quote this coefficient.

The predictions for the quiver at large p follow. The entropy and the energy per fermion are universal functions of βJ (Figure 6a,b), with $S_0/N = \ln \sqrt{3}$, the value already found for the index in Section 6, and at low temperature $E/N = 0.1667 T^2/J$ and $S/N = \ln \sqrt{3} + 4\pi^2 \alpha_s T/J$. The Schwarzian coefficient is $C = N\alpha_s/J$. Turiaci and Witten give the threshold of a non-BPS multiplet with charges $q \pm \hat{q}/2$ as $E_0(q) = q^2/4\hat{q}^2$ in units of $1/2C$ [4], so

$$E_0(q) = \frac{Jq^2}{8\hat{q}^2 \alpha_s N} = 1.645 \frac{Jq^2}{N} = 0.548 q^2 \frac{p\langle C^2 \rangle}{n}, \quad (18)$$

and the semiclassical entropy of [3] gives the same threshold independently. The saddle also says when (18) can be tested. The energy $E_0(q)$ corresponds to a temperature $T \approx 3.1 Jq/N$, and there the first correction to the T^2 law, $a_3 T/a_2 = -1.56 T/J$, is about 30% for $q = 1.5$ at $N = 24$ and still 12% at $N = 60$. A quantitative test therefore needs $N \gtrsim 60$ –100 fermions.

8.2 Exact tests at $n = 1$

The $n = 1$ model is tripartite $\mathcal{N} = 2$ SYK with $3p$ fermions and $Q = \sum C_{fgh} a_f^\dagger b_g^\dagger c_h^\dagger$. Without the gauge projection it can be diagonalised completely up to $p = 6$ (2^{18} states, block by block in the three fermion numbers), and its second moment can be computed exactly. Because $Q^2 = 0$, cyclicity gives $\langle H^2 \rangle = 2\langle QQ^\dagger QQ^\dagger \rangle$. Of the three Wick pairings of the Gaussian couplings, two give $(p^2 + p)^3/64$, from products of the number operators $T_I T_I^\dagger = n_{af} n_{bg} n_{ch}$ with $T_I = a_f^\dagger b_g^\dagger c_h^\dagger$, and the third gives only $p^3/8$. With $\sigma^2 = \langle C^2 \rangle$ and $\mathbb{E}\langle H \rangle^2 = \sigma^4(p^6 + 2p^3)/16$ this yields

$$\frac{\mathbb{E} \text{Var}(H)}{N} = \frac{J^2}{16} \left(1 + \frac{1}{p} + \frac{1}{p^2} \right), \quad (19)$$

p	sectors	BPS states	sectors with $\#BPS = \chi $	BPS states at $J_R = -1, 0, 1$ (or $\pm\frac{1}{2}$)	outside $ J_R < \frac{3}{2}$
2	19	36	19	9 : 18 : 9	0
3	37	172	36	81 : 81	10
4	61	984	61	246 : 486 : 246	6
5	91	4536	91	2 190 : 2 190	156
6	127	26 400	127	6 597 : 13 128 : 6 597	78

Table 8: BPS states of the full $n = 1$ model by gauge-charge sector, from exact diagonalisation (twenty coupling draws at $p \leq 5$ and three at $p = 6$, all identical). Wherever $\#BPS = |\chi|$ the states occupy a single block. The exceptional sector at $p = 3$ is the singlet sector, with $\chi = 0$ and $2 + 2$ states at $J_R = \pm\frac{3}{2}$.

whose limit $J^2/16$ is exactly the slope of the saddle. The second moment therefore approaches the saddle with corrections $1/p + 1/p^2$, and twenty coupling draws at each $p \leq 5$ agree with (19) within their errors (Figure 6c). The full thermodynamics behaves in the same way. For $\beta J \lesssim 5$ the deviations from the saddle fall off as $1/p$: $p\Delta S/N = -0.0072, -0.0075$ and -0.0072 at $\beta J = 1$, and $p(E/E_{SD} - 1) = -0.97, -1.03$ and -0.99 at $\beta J = 2$, for $p = 4, 5$ and 6 . At low temperature the exact entropy saturates at $\ln(\#BPS)/N$, which lies above $\ln\sqrt{3}$ by a term that we identify below.

8.3 Concentration in every gauge-charge sector

The exact $n = 1$ spectra contain much more than the singlets. Since Q raises each of the three fermion numbers (x, y, z) by one, $(s, t) = (x - y, y - z)$ is conserved. These are the $U(1)^3$ gauge charges, and the singlet sector is $(0, 0)$. Each sector is a chain of blocks whose R-charge $J_R = x + y + z - \frac{3}{2}p$ runs over a single residue class mod 3, and its index is $\chi(s, t) = \sum (-1)^{x+y+z} \dim(x, y, z)$. If each sector were a generic $\mathcal{N} = 2$ theory obeying the Turiaci–Witten window $|J_R| < \hat{q}/2$, its BPS states would all sit at its unique charge inside the window, and their number would be $|\chi(s, t)|$. Table 8 shows that this is exactly what happens. In every sector at $p = 2, 4, 5$ and 6 , and in every sector but one at $p = 3$, the number of BPS states equals $|\chi|$ and the states occupy a single block. The exception is the singlet sector at $p = 3$, whose $2 + 2$ states at $J_R = \pm\frac{3}{2}$ (Section 7.7) sit on the edge of the window. The counts are the same for every coupling draw. Summing $|\chi|$ over sectors, which can be done exactly at any p , gives $\frac{4}{3} 3^{N/2}$ BPS states for even N , in the ratio 1:2:1 at $J_R = -1, 0, 1$, and $\frac{2}{\sqrt{3}} 3^{N/2}$ split equally between $J_R = \pm\frac{1}{2}$ for odd N , up to a remainder outside the window that disappears quickly for even N and slowly for odd N (Figure 6d,e). These are precisely the exact ground-state counts of $\mathcal{N} = 2$ SYK, $D(N, 0) = 2 \cdot 3^{N/2-1}$ and $D(N, \pm\frac{1}{3}) = 3^{N/2-1}$ for even N [3], and the cosine law of [4]. The low-temperature entropy offset of the previous subsection is the prefactor of these counts, $\ln(\sum |\chi|/3^{N/2})/N \rightarrow \ln\frac{4}{3}/N$, which equals 0.0163 at $p = 6$. At $n = 1$, then, R-charge concentration is not a property of the singlets alone. It holds in every gauge-charge sector, and the window is that of $\mathcal{N} = 2$ SYK.

This suggests a rule for the singlet sector at any n . Singlet BPS states are the invariant part of the cohomology of the ungauged model, so if the ungauged model obeys the window sector by sector, as the $n = 1$ model does, the singlet BPS states lie at $|J_R| < \frac{3}{2}$. Singlet degrees are multiples of three, so $J_R \in 3\mathbb{Z}$ when pn^2 is even and $J_R \in \frac{3}{2} + 3\mathbb{Z}$ when it is odd. For even pn^2 the rule therefore says that all singlet BPS states sit at half filling, $|J_0|$ of them, which is the concentration found at $(2, 2)$, $(2, 3)$ and $(3, 2)$ and at even p for $n = 1$. For odd pn^2 it says that there are no singlet BPS states at leading order, consistent with the vanishing of I_0 (Section 4), with $n = 1$ at $p = 5$, and with $p = 7$ outside its central blocks; the only exception we know is the window-edge pair at $(1, 3)$. For $(3, 3)$, where $pn^2 = 27$, the rule predicts no singlet BPS states at all, or at most a few in the two central degrees $k = 39$ and 42 . It is a heuristic: it rests on the large- e^{S_0} statement of [4], and at $n \geq 2$ only the singlet data support it.

8.4 Near-BPS edges

Equation (18) can be confronted with the lowest singlet multiplets at $n = 1$, which we computed up to $p = 10$ for Gaussian couplings (Table 9, Figure 6f). The overall scale is right: for $N \geq 12$ every measured edge is within a factor 2.6 of the prediction. The edges have not converged, however. At fixed q the ratio of measured to predicted still drifts downward with p , from 1.09 to 0.38 at $q = 1.5$ between $p = 4$ and 8 , and the edges at small q lie below the threshold, which the finite- e^{S_0} fluctuations of a random-matrix edge cannot produce, since they raise the lowest level. With $N \leq 30$ the data are pre-asymptotic, as the estimate of the previous subsection anticipates.

The integer couplings used for the exact and bootstrap computations at $n \geq 2$ need a separate remark. They

p (N)	J	$q = 1.5$	$q = 4.5$	$q = 7.5$	$q = 10.5$
4 (12)	0.736	0.246 (1.09)	2.95 (1.44)		
6 (18)	0.878	0.112 (0.62)	1.68 (1.04)	5.27 (1.17)	
8 (24)	0.847	0.049 (0.38)	0.85 (0.72)	3.05 (0.93)	6.77 (1.06)
10 (30)	0.976		0.73 (0.68)	2.39 (0.79)	

Table 9: Lowest singlet multiplet energies $E_0(q)$ at $n = 1$ (Gaussian couplings, one draw, realised $J = p^2 \langle C^2 \rangle$), with the ratio to the large- p prediction (18) in parentheses. For odd p the ratios are 0.71 and 1.27 at $q = 3$ and 6 for $p = 5$, and 0.62, 0.88 and 1.10 at $q = 3, 6$ and 9 for $p = 7$.

property	$p = 2$	$p = 3$ (and $p \geq 3$)
BPS entropy	$\ln I_0 = 3n \ln 3 + O(\ln n)$, proved; not macroscopic	$\ln I_0 = F^*(p)n^2 + O(\ln n)$, $F^*(3) = 1.2373$, 41% of the singlet log-count (large-deviation level; sign proviso for odd p)
Concentration	(2, 2): all 90 singlet BPS states at $k = 12$, exact; (3, 2): all 1680 at $k = 27$, numerical	(2, 3): all 1680 at $k = 18$, numerical; (3, 3): none at $k \leq 21$ (bootstrap), open for $24 \leq k \leq 36$
Fortuity (rank)	(2, 2): all 90 classes, to all orders	(2, 3): all 1680 classes (exact certificate); (3, 3) classes at $k = 39$; (2, 4) classes at half filling
Non-BPS statistics	not random-matrix; hidden $\mathcal{N} = 4$ on singlets	Gaussian orthogonal, 10 024 levels at (2, 3)
Near-BPS gap	none (≥ 95.6)	0.054 next to the BPS degree at (2, 3) (integer couplings); large- p threshold $E_0(q) = 1.645 J q^2 / N$, met within a factor 2.6 at $n = 1$, $N \leq 30$, not converged; a test needs $N \gtrsim 60$
Supersymmetry	$\mathcal{N} = 4$ on singlets (numerical, $n = 2, 3$)	$\mathcal{N} = 2$
Large n	planar; exact index	planar; index saddle solved
Large p	n/a	$\mathcal{N} = 2$ SYK with $\hat{q} = 3$, saddle solved ($\alpha_s = 0.00844$); at $n = 1$, $p \leq 6$, every gauge-charge sector concentrated, with the SYK charge distribution

Table 10: Desirable properties of the quiver for $p = 2$ and $p \geq 3$, with their status.

have mean 3 and variance 2, and the mean part of the supercharge, $3p^{3/2} \text{Tr}(\bar{A}\bar{B}\bar{C})$ in terms of flavour-averaged fields, acts coherently on the $3n^2$ flavour-symmetric modes with a weight comparable to that of the random part. The energy fluctuations are then not extensive. At $n = 1$, $\text{Var}(H)/NJ^2 = 0.106, 0.146, 0.190$ and 0.236 for $p = 2$ to 5, growing linearly in p instead of approaching $1/16$. The integer ensemble is therefore outside the large- p SYK class, (18) does not apply to it, and nothing guarantees that its spectra follow the Turiaci–Witten law at all. This is a further reason why the (2, 3) edges of Section 5 do not test the super-Schwarzian; a Gaussian-coupling computation at (2, 3) would be needed. Cohomological statements are unaffected, since they hold for generic couplings, and indeed the BPS counts of Table 8 are the same in both ensembles.

9 Prospects for the $p \geq 3$ quiver

Table 10 sets the two cases side by side. The contrast is sharp. The $p = 2$ quiver is an elegant and highly structured corner, with a hidden $\mathcal{N} = 4$ on singlets, an exact index and a spectrum that is not random-matrix like, but its BPS entropy is provably sub-macroscopic. The $p \geq 3$ quiver has every property we have been able to test at the level we would hope for: a singlet BPS entropy that is a finite fraction of the total at order n^2 , a concentrated BPS sector, chaotic non-BPS spectra, small gaps next to the BPS degree, and only $\mathcal{N} = 2$ supersymmetry, which is what the $\mathcal{N} = 2$ super-Schwarzian needs. The large- p limit adds a pleasing interpolation: the singlet BPS entropy per fermion rises from zero at $p = 2$ towards the SYK value at $p = \infty$, so the flavour number dials between a gauge-dominated matrix regime and an SYK-like one, with planar large- n control at every p .

The caveats are equally clear, and they define the programme. Much of this rests on $n = 2$, and concentration is the property most at risk as n grows: in the adjoint models it was the growth of the window with the rank that failed, and although its mechanism is absent here, nothing yet proves that another one does not take its place. At $p = 2$ concentration survives the step to $n = 3$. At $p = 3$ the bootstrap confines the rank-3 singlet BPS states to

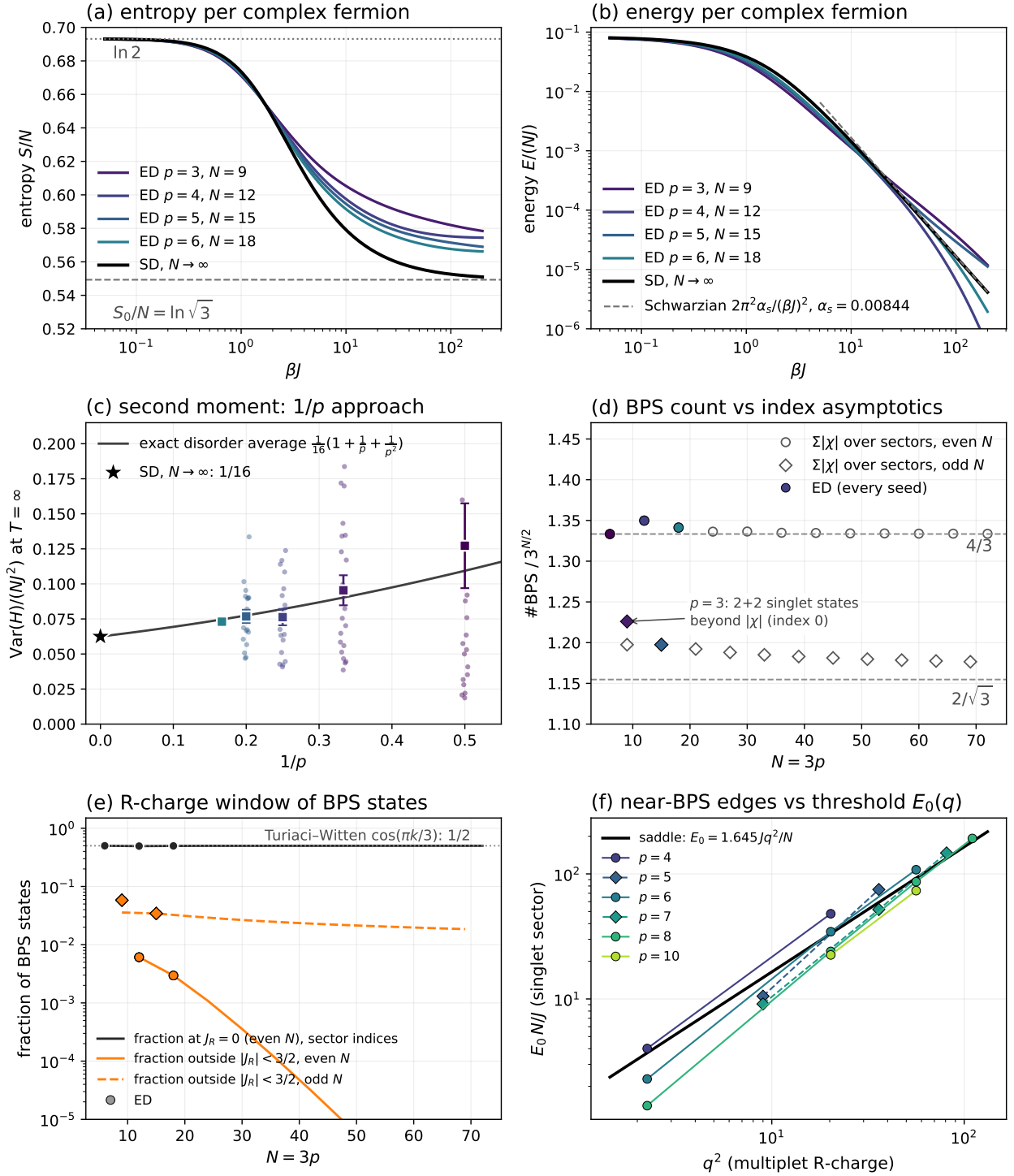


Figure 6: The large- p saddle against the exactly diagonalised $n = 1$ model. (a) Entropy and (b) energy per complex fermion against βJ , exact at $p = 3$ – 6 (averaged over coupling draws) and from the saddle, with $\ln 2$, $S_0/N = \ln \sqrt{3}$ and the Schwarzian asymptote. (c) Second moment of H at infinite temperature against $1/p$: coupling draws (dots), their means with standard errors (squares), the exact average (19) and the saddle value $1/16$ (star). (d) Number of BPS states over $3^{N/2}$: exact diagonalisation (filled) and the exact sum of sector indices (open), with the $\mathcal{N} = 2$ SYK values $\frac{4}{3}$ and $\frac{2}{\sqrt{3}}$. (e) Fraction of BPS states at $J_R = 0$ for even N , and outside the window $|J_R| < \frac{3}{2}$, from the sector indices (lines) and exact diagonalisation (markers). (f) Lowest singlet multiplet energies against q^2 , with the threshold (18).

$24 \leq k \leq 57$, which is far from a proof, because the open degrees are precisely those nearest half filling, where a widening window would first appear. The near-BPS gaps at $(2, 3)$ are suggestive, but at this size the neighbouring charge sectors differ too much in entropy for the Turiaci–Witten density to apply, and the integer couplings lie outside the large- p SYK class. The large- p saddle now predicts the thresholds, but testing them needs $N \gtrsim 60$ fermions, so the super-Schwarzian question remains open. Fortuity at $p = 3$ is now established through the $(3, 3)$ bootstrap, with certificates verified in exact arithmetic. And the model is planar rather than melonic, so at fixed p there is no analogue of the SYK Schwinger–Dyson solution; large- n dynamics has to come from the bootstrap, from other matrix-model methods, or from the large- p limit, whose saddle is solved in Section 8.

Several next steps follow, and the first observation is that concentration beyond $n = 2$ has to be settled without help from the index. Since Q raises the fermion number by three and every singlet has $k \equiv 0 \pmod 3$, no R-charge fugacity is compatible with the pairing of non-BPS states in the singlet sector: $\text{Tr}(-1)^k y^{k/3}$ is protected only at $y = 1$. The singlet index is therefore the only protected quantity of the gauged model, and its sign, which always agrees with half filling, is all it can say about where the BPS states sit.

The exact methods have now reached their limit. The Schur–Weyl basis of Section 7 has delivered the $(3, 2)$ point, where the near-BPS gap and the multiplet statistics can also be followed from $n = 2$ to $n = 3$, albeit in the $p = 2$ corner. At $(2, 3)$, with 300 584 singlets at half filling, it can still supply the 1680 BPS vectors for a direct test of BPS chaos at $p = 3$. Beyond that the singlet sectors explode: at $(3, 3)$ they hold 1.2×10^8 states at $k = 18$ and 1.1×10^{12} at half filling, and 3×10^8 at half filling at $(2, 4)$.

At $p = 3$ and $n \geq 3$ the matrix bootstrap [15, 16, 17] has taken over, in the singlet form of Section 7. It has delivered its first target, the certificate at $(3, 3)$ and $k = 18$ that makes the $(2, 3)$ classes fortuitous, now verified in exact arithmetic, and it has mapped its own frontier, which settles near filling 0.27 as the rank grows. The second target, certificates for $24 \leq k \leq 36$, which would confine the rank-3 BPS states to the two singlet degrees adjacent to half filling, is out of reach at level 2, and the diagnosis of Section 7 indicates that more operators alone will not reach it. These are also the only degrees in which the window rule of Section 8 allows any singlet BPS states at $(3, 3)$, and it expects few or none. What is needed are constraints that tie positivity to the gauge representation of the state more tightly than Casimir conditions do; whether Gram matrices of products of open words, resolved into gauge irreps, or a formulation that works inside the singlet sector from the start can achieve this is the next methodological question. The near-BPS gaps and their scaling with n , the actual super-Schwarzian test, remain the harder final target.

On the analytic side, the large- p limit is now under quantitative control (Section 8). The saddle fixes the super-Schwarzian coefficient and the near-BPS thresholds, and at $n = 1$ every exact check agrees with it. Three things remain. The chaos exponent should be computed from the ladder kernel of the same saddle. The dominance of melonic diagrams with couplings shared by n^3 colour triples should be checked, for instance through the $1/p$ coefficient of $\text{Var}(H)$ at $n \geq 2$ against its value $1/p + 1/p^2$ at $n = 1$. And the drift of the $n = 1$ edges should be compared with standard $\mathcal{N} = 2$ SYK at the same N , to decide whether it is generic finite- N behaviour. A genuine test of (18) needs $N \gtrsim 60$, beyond exact methods, which makes it a natural target for a bootstrap organised around large p rather than large n . That $F^*(p)$ is smooth for all $p > 2$ suggests that $p = 3$ lies in the same phase. Monte Carlo integration of (8), by thermodynamic integration in p from the exact point $p = 2$, would test the finite- n predictions at $n = 6$ –12, and the generalised monotones of [14] should be checked. Finally, the hidden $\mathcal{N} = 4$ of the $p = 2$ quiver deserves a derivation and a test at $n = 3$, both for its own sake and because a small $\mathcal{N} = 4$ super-Schwarzian would require an $SU(2)$ R-symmetry that we have not found.

In summary, the fermionic $U(n)^3$ quiver with $p \geq 3$ flavours is, to our knowledge, a new candidate for a matrix SYK model. It has macroscopic and, at the sizes we can reach, concentrated BPS entropy, chaotic multiplet spectra and near-BPS gaps; its rank-2 BPS classes are fortuitous at both $p = 2$ and $p = 3$; it reduces to an exactly solvable log gas at the level of the index; and its large- p limit is $\mathcal{N} = 2$ SYK, whose saddle we have solved and which the exactly diagonalised $n = 1$ model matches, both in its thermodynamics and in the R-charge distribution of the BPS states of every gauge-charge sector. Whether concentration survives at large n is now the central question. It holds at $(3, 2)$, and at $(3, 3)$ the bootstrap excludes singlet BPS states from every degree except the twelve nearest half filling; sharper bootstrap constraints for those degrees and a test of the near-BPS thresholds beyond exact methods are the next steps.

A Methods and reproducibility

All computations are reproducible from the repository, with the parameters, seeds, outputs and memory use recorded in the result files. Cohomology at $(2, 2)$ uses exact ranks modulo primes. The $n = 2$ spectra use Weyl-

reduced zero-weight sectors (`src/quiver_n2.py`) with the exact Casimir penalty, Lanczos (`scripts/quiver_singlet_zero_modes.py`), dense singlet bases (`scripts/quiver_singlet_spectrum.py`), the Casimir-polynomial projection for the 10024-level pair (`scripts/quiver_singlet_pair_large.py`, float64, peak 8.6 GB) and the $\mathcal{N} = 4$ and symmetry analysis (`scripts/quiver_p2_n4.py`, `scripts/quiver_commutant.py`). The fortuity test is `scripts/run_quiver_fortuity.py`. The Schur–Weyl singlet basis is `src/quiver_singlet_basis.py` with `scripts/quiver_singlet_spectrum_sw.py`, tested by `tests/test_quiver_singlet_basis.py` (singlet dimensions, $Q^2 = 0$, and the stored $n = 2$ spectra). The bootstrap uses the quiver trace algebra `src/quiver_trace.py`, checked against explicit Fock-space operators by `tests/test_quiver_trace.py`, and the semidefinite program `src/quiver_bootstrap.py` with `scripts/quiver_bootstrap_run.py`; `tests/test_quiver_bootstrap.py` evaluates every constraint on exact functionals. The full-sector zero modes are `scripts/quiver_sector_zero_modes.py`, and the bootstrap figure is `scripts/plot_quiver_bootstrap.py`. Exact certificate verification is `src/quiver_certificate.py` with `scripts/quiver_verify_certificate.py`, the frontier scans `scripts/quiver_bootstrap_frontier.py`, and the $n = 1$ uplift channels `scripts/quiver_n1_arrow_tower.py`. The large- p saddle is `src/n2syk_sd.py` with `scripts/n2syk_thermo.py` (midpoint grid with $\beta J/M \leq 0.002$, Richardson extrapolation, spline quadrature), tested by `tests/test_n2syk.py` (the three Wick pairings from explicit Fock operators, the high-temperature limits and the conformal limit); the full $n = 1$ diagonalisation is `scripts/quiver_n1_full_thermo.py` (dense blocks, zero modes below 10^{-12} , peak 3.5 GB at $p = 6$), and Figure 6 is `scripts/plot_n2syk_large_p.py`. Singlet indices use `src/quiver_index.py` (dual Cauchy and the Littlewood–Richardson transfer matrix, exact integers). The log-gas analysis is `src/quiver_saddle.py` and `scripts/quiver_index_saddle.py`, tested by `tests/test_quiver_saddle.py` (reduced against full energies, Hessians against finite differences to 10^{-10} , the $p = 2$ formula, the continuum values, and the eigenvalue representation against direct quadrature at $n = 1$). The derivations are written out in `docs/derivations.md` (D18 to D22) and the research log in `research/notes/quiver_project.md` (Sections 11 and 12 for the material of Section 7, Section 14 for Section 8). The main data files are `results/data/quiver_singlet_index_dixon.json`, `quiver_singlet_zero_modes_n2_p3_k*.json`, `quiver_singlet_pair_n2_p3_k9_f64.json`, `quiver_p2_n4*.json`, `quiver_fortuity_n2_p2_k12.json`, `quiver_index_saddle.json`, `quiver_singlet_sw_n3_p2_seed3.json`, `quiver_singlet_sw_n3_p3_seed3.json`, `quiver_bootstrap_n*_p*_L*.jsonl`, `quiver_bootstrap_frontier_diagnostics.txt`, `quiver_bootstrap_frontier_n*_p*.jsonl`, `quiver_certificate_checks.jsonl` with the certificates `quiver_certificate*.npz`, `quiver_n1_arrow_tower_pmax5.json`, `n2syk_thermo_qh3.json`, `quiver_n1_full_thermo.jsonl`, `quiver_n1_full_moments.jsonl`, `quiver_n1_full_moments_int.jsonl` and `quiver_n1_spectrum_gauss_seed1*.json`. The bootstrap runs used the seed-3 integer couplings and the interior-point solver Clarabel 0.11.1 (SCS 3.3.1 for the independent certificate), with peak memory 3–9.7 GB.

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